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Enhancing Web Application and run tests
on simulations for Energy Market

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MAON



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Dedication

I couldn't have accomplished this work without the incredible people who believed in me and placed their trust in my abilities. Their faith gave me the strength to push through challenges, and words can hardly express the love and gratitude I feel for all of them. I truly hope that one day I'll be able to return the kindness, support, and encouragement they've given me throughout this journey.

Family always comes first. You are the foundation of everything I am today and the reason behind every success I've achieved. I've been blessed with a caring family who has stood by my side through both good and occasionally hard times, always keeping me grounded and making sure I was doing well, even while I was miles away in Berlin. Mom, Dad, my brother and my sister, you are my greatest source of strength and love. And of course, Seif and Monica, you're family too, and I truly appreciate the warmth and positivity you've shown me.

To my friends and colleagues, thank you for being such an incredible part of this experience. You made my six-month internship fly by in what felt like a blink of an eye, filling it with laughter, learning, and unforgettable memories. And to my childhood friends—you know who you are—it always feels like no time has passed when we share happy moments together. Even from afar, you made me feel like I never left Tunisia. I'm truly lucky to have you all in my life, and I hope our friendship continues to grow for years to come.

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Finally, I would like to thank all those who, directly or indirectly, contributed to the success of this internship — your encouragement, trust, and assistance made this experience a truly rewarding one.

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General Introduction

As part of my final year project, I was given the opportunity to collaborate with Maon GmbH, a Berlin-based company in Germany. A 6-month internship was established to develop, maintain, and implement innovative features in Maon's existing solution. The technology stack consisted of a fullstack architecture based on the MERN framework (MongoDB, ExpressJS, ReactJS, NodeJS), complemented by related technologies including GraphQL with Apollo server, MinIO, Bootstrap CSS, Python, and other supporting tools.

The internship encompassed multiple ongoing projects, which can be categorized into two main initiatives:

- Project 1: Maon web application: Main platform that contains Maon's services which supports users collaborating on projects, scenarios, and simulations.
- Project 2: Test Maon simulation app: Allows developers to test the core applications of Maon using dynamically generated inputs and outlaying outputs.

In this report, we've talked about problem statement and analysis of existing solution along with mentioning the hosting company and the proposed solution that brought fixes to the platform. Then we discussed the planification of the project and the teamwork behind it without forgetting the multiple architectures used to understand the environment in a better way. For the last two chapters, we discussed the conception and different utilization of the web application, the realization and technologies used to obtain an efficient result.

This period offered valuable opportunities for learning, growth, and professional development in a supportive environment. This document explores and goes into deep details of the projects completed and the work that contributed to their success.

Chapter 1: Problem statement and analysis of existing solution

I. Introduction

Renewable energy [1] has emerged as one of the fastest-growing markets in Europe over the past two decades, attracting significant investment from companies and entrepreneurs seeking profitable opportunities. Maon focuses specifically on the electricity market, where energy engineers execute simulations to forecast market trends and identify profitable gaps that help investors maximize returns. In this context, a web application is provided to deliver these services and additional features to clients.

II. Academic Environment

This project was completed as part of my graduation requirements for the Software Architectural Engineering degree at the Private Higher School of Engineering and Technology (ESPRIT). During a 6-month internship at Maon GmbH in Berlin, Germany, I maintained and enhanced Maon's web services for energy forecasting. Also, I worked on setting up comprehensive testing procedures. In my role as Software Developer intern, I worked on two primary projects that were connected eventually and resolved numerous issues to build a more responsive and efficient web application, ultimately improving Maon's core platform.

III. Hosting company

The host organization for this project is Maon GmbH, a Berlin-based company specializing in fundamental electricity market simulations [2] and forecasting. Its mission is to revolutionize decision-making in energy markets by providing unparalleled foresight—exceptional and unmatched insights that anticipate future developments, trends, and requirements in the energy sector. The company provides users with a web application that displays available services and simplifies participation and investment in the energy market.



Figure 1: Hosting company Logo

IV. Study Of Existence

1. Troubleshooting

Maon's energy market solution may be available and it is providing its customers with the best functionalities in the market yet. We can mention that it offers a lot of services as shown in the next table:

Table 1: Table of existing functionalities and errors in the current Maon web application

Category	Functionality	Developer Troubleshooting
Energy Monitoring	Real-time data visualization and reporting	Encountered occasional data delays and chart rendering issues.
Device Management	Device registration and configuration	Experienced backend registration errors.
Alert System	Automatic alerts and notifications	Alerts were not triggering, and email delivery was failing.
Authentication	Secure login and user management	Token expiration and session handling issues were causing user logouts.
Frontend–Backend Integration	Data synchronization via APIs	API endpoint mismatches and CORS errors occurred during integration.
CI/CD Pipeline	Jenkins build and deployment	Maven and Nexus deployment errors interrupted the CI/CD pipeline.
UI/UX Stability	Responsive dashboard and data filtering	Layout bugs appeared, and loading states were missing in some views.
Documentation & Logging	API and system documentation	Missing logs and unclear API usage caused debugging difficulties.

2. Proposed solution and fixes

The internship focused on enhancing existing features and developing new functionality in the Maon application. Part of the feasibility study examined the current application, identifying buggy features and determining necessary improvements to create a better user experience and attract more customers. It also focused on Maon simulation applications and worked on improving them with testing and analyzing.

V. Company's workflow

This project was primarily managed using SCRUM methodology. This approach facilitated effective project structure organization, streamlined teamwork functions, and established clear responsibilities. The methodology enabled developers, project stakeholders, and clients to access all necessary project information efficiently, resulting in faster delivery and improved efficiency. We can see from the next figure a detailed structure of the SCRUM methodology:

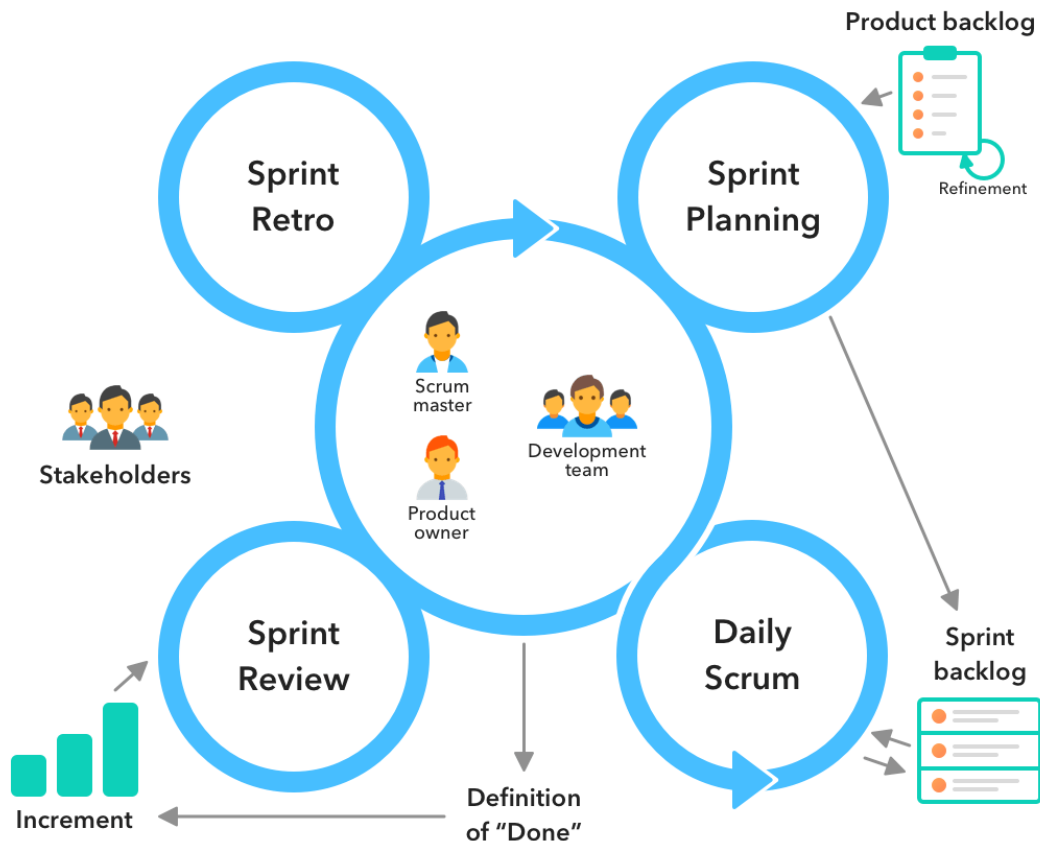


Figure 2: Scrum methodology and concept [3]

The Scrum methodology comprises the following components:

- **Product Owner:** Responsible for defining product vision, maintaining and prioritizing the product backlog, and ensuring that development efforts align with business objectives and customer needs. The Product Owner serves as the primary liaison between stakeholders and the development team.
- **Scrum Master:** Facilitates Scrum ceremonies, removes impediments that hinder team productivity, ensures adherence to Scrum principles, and fosters a collaborative environment that promotes continuous improvement.

- **Sprints:** Time-boxed iterations, typically lasting 2-4 weeks, during which specific backlog items are developed, tested, and delivered. Each sprint produces a potentially shippable product increment.
- **Daily Scrum:** Brief daily synchronization meetings where team members share progress updates, discuss planned work for the day, and identify any blockers requiring immediate attention.
- **Sprint Review (Revue de Sprint):** Conducted at the end of each sprint to demonstrate completed work to stakeholders, gather feedback, and assess whether sprint goals were achieved.
- **Sprint Retrospective (Rétrospective de Sprint):** A reflection session held after each sprint where the team analyzes what worked well, identifies areas for improvement, and defines actionable steps to enhance future sprint performance.
- **Stakeholders:** individuals or groups with an interest in the product who provide feedback and requirements represented by the Product Owner.
- **Sprint Backlog:** list of selected Product Backlog items and tasks that the Scrum Team commits to complete during a specific Sprint.

VI. Conclusion

This project has been operational for some time, competing with multiple market leaders while offering enhanced functionalities and practical services that increase customer satisfaction and encourage greater investment. The application has undergone multiple version iterations, each addressing bugs from previous releases. My responsibility was to enhance and maintain these features while contributing to ongoing development. This work was supported by experienced mentors and developers working within a well-organized structure facilitated by Scrum methodology.

Chapter 2: Planification (Sprint 0)

I. Introduction

This chapter represents a critical component in the project development process. It enabled us to clearly define our objectives for the deliverables scheduled during the 6-month period, ensuring streamlined and efficient workflow execution. The chapter encompasses backlog preparation, identification of actors who directly utilize the services, and comprehensive workload planning.

II. Requirements capture

For any company, customers establish key standards and require ongoing adjustments to ensure long-term satisfaction and retention. This necessitates alignment between company capabilities and customer requirements to guarantee project success and achieve customer satisfaction when deliverables meet or exceed expectations.

1. Actors Identification

Both direct and indirect actors are paramount stakeholders, as they represent our target audience and must be satisfied with the services Maon provides. The primary actors include:

- **Private Investors:** Individual and institutional investors seeking profitable opportunities in the energy market rely on accurate forecasting and market gap analysis. They use Maon's services to make informed investment decisions, optimize portfolio performance, and identify emerging opportunities in the electricity sector.
- **Energy Companies:** Utility providers, power generation companies, and energy trading firms constitute the core user base. They depend on simulation services for capacity planning, risk assessment, price forecasting, and strategic market positioning to maintain competitive advantages.
- **Energy Consultants:** Professional advisors and consultancy firms that provide strategic guidance to clients in the energy sector utilize the platform's analytical capabilities to deliver data-driven recommendations.
- **Research Institutions:** Academic and research organizations studying energy market dynamics, sustainability initiatives, and grid optimization leverage the simulation tools for analysis and model validation.

2. Functional requirements

The functional requirements for this project addressed numerous bugs and system malfunctions requiring algorithmic enhancements to ensure smooth operation and optimal efficiency for customers. Key requirements included:

- **Update Components' Map:** Enhance the geographical visualization system to accurately display energy infrastructure components, including power plants, transmission lines, and distribution networks. This involved fixing marker placement issues, improving layer rendering, and implementing real-time updates to reflect current system configurations.

- **Fix scenario year creation:** enhance scenario inheritance methodology, keeping the parents and children in synchronization. This inheritance is crucial for defining intervals in scenarios and using it as a future need.
- **Modify Sign-up Process:** Redesign the user registration workflow to streamline the account creation process, implement proper validation mechanisms, and enhance security protocols. This included fixing the form submission errors, improving the user experience, and integrating email verification systems.
- **Fix Imports/Exports:** Resolve data import and export functionality issues to ensure seamless data exchange between the platform and external systems. This involved correcting the file format compatibility, fixing parsing errors, and implementing robust error handling for large datasets.
- **Bulk Handling Algorithms:** Develop and optimize algorithms for processing multiple operations simultaneously, enabling users to manage large-scale data manipulations efficiently. This included implementing batch processing capabilities and improving the system performance during concurrent operations.
- **Test Simulations Faster:** Optimize simulation execution times through code refactoring, algorithm efficiency improvements, and implementation of parallel processing techniques to reduce computational overhead and deliver results more rapidly.

3. Non-functional requirements

Non-functional requirements are equally essential components in project development, addressing critical aspects that contribute to building a robust and resilient web application:

- **Security:** Implement comprehensive security measures including data encryption (both in transit and at rest), secure authentication mechanisms, role-based access control and handle other vulnerabilities that could face us.
- **Scalability:** Design the system architecture to accommodate growing user bases and increasing data volumes without performance degradation. This includes implementing horizontal scaling capabilities, database optimization, and load balancing mechanisms.
- **Performance:** Maintain optimal response times, minimize latency, and ensure efficient resource utilization. This involves implementing caching strategies, optimizing database queries, and monitoring system performance metrics continuously.
- **Compliance:** Adhere to industry standards, regulatory requirements, and best practices in energy market operations. This includes maintaining documentation, ensuring audit readiness, and implementing processes that align with international standards.
- **Maintainability:** Structure code and documentation to facilitate easy updates, bug fixes, and feature additions by current and future development teams.
- **Reliability:** Ensure high availability and system uptime through redundancy measures, failover mechanisms, and comprehensive error handling.

4. Logical architecture

The web application is based on Mern stack framework. This means that logical architecture is a 3-tier architecture. The Model-View-Controller (MVC) [4] architecture is a well-established design pattern widely adopted in MERN stack projects. This architectural pattern consists of three fundamental components:

- **Model:** Represents the data layer and business logic, managing database interactions through MongoDB [5] schemas and handling data validation, storage, and retrieval operations.
- **View:** Constitutes the presentation layer built with ReactJS [6], responsible for rendering the user interface and displaying data to end users in an intuitive and accessible format.
- **Controller:** Implements the application logic layer using ExpressJS [7] and NodeJS, processing user requests, coordinating between models and views, and managing the application workflow.

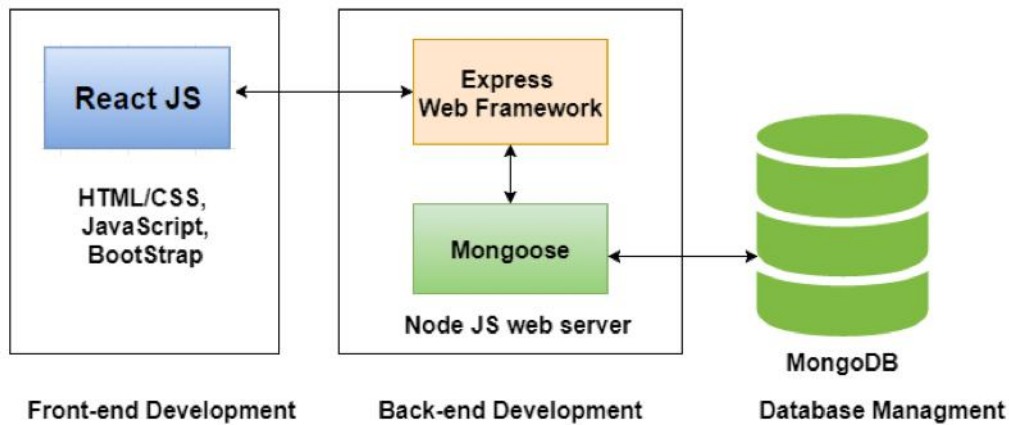


Figure 3: 3-tier MVC architecture used for Maon's Web Application

5. Physical architecture

This physical architecture represents an MVC-based MERN deployment on a Kubernetes cluster. The Front-End pod (View) hosts the React app, the Back-End pods (Controller) run Node.js [6] and Apollo Server to handle logic and API calls, and the MongoDB database (Model) manages data storage. Telepresence enables developers to mirror and intercept live traffic for debugging, while Ambassador manages external routing to the cluster, ensuring efficient communication and scalable development.

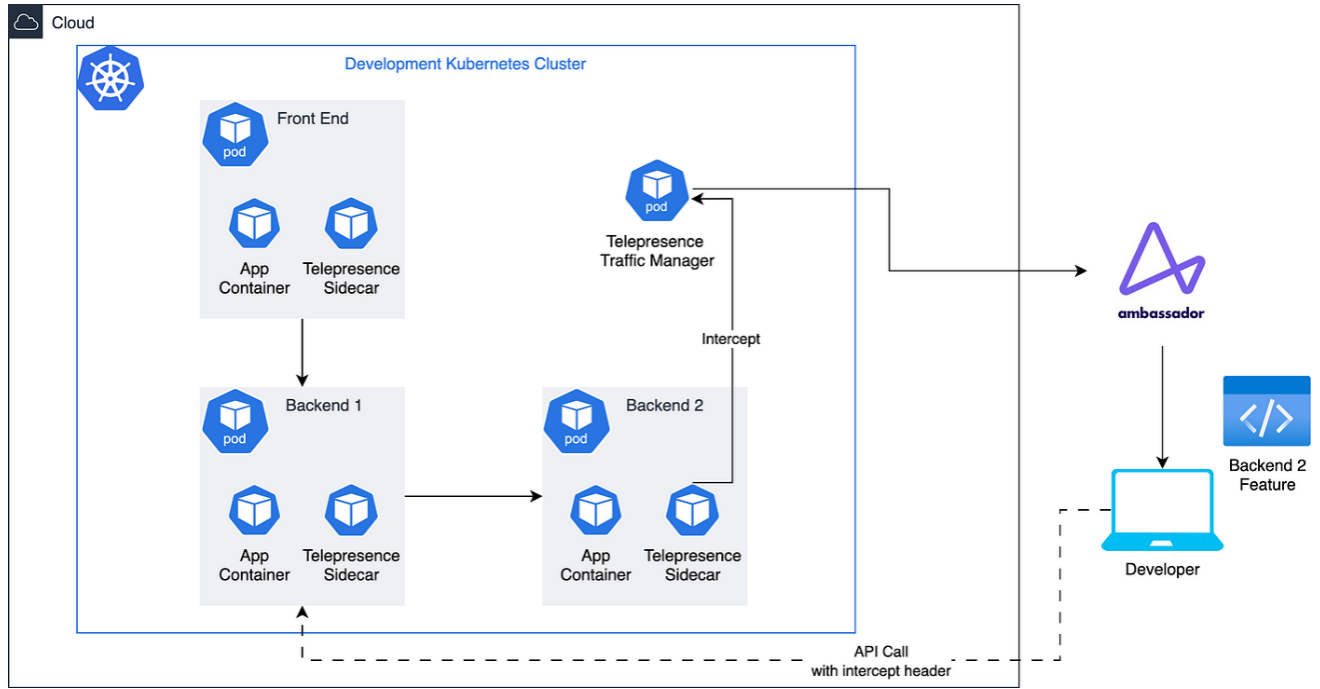


Figure 4: Physical architecture in development process

6. Scrum implementation

a. Roles :

- Product Owner: Mihail Ketov (Co-Founder and Managing Director), Yash Petal (Senior Simulation Engineer)
- Scrum Master: Fabian Breitreutz (Co-Founder and Development Head)
- Developers: Bacem Jaza, Hajer Mouchrak, Reinis Muiznieks, Sangeetha K.

b. Product backlog :

The product backlog contains numerous issues and feature enhancements identified during this internship to improve project functionality and overall composition. Table 2 outlines the different tasks categorized by user role:

- **Admin:** System configuration, user management, access control, platform monitoring, data integrity maintenance, and administrative reporting.
- **User/Investor/Company:** imports/exports customization, simulation execution, report generation, market analysis access, notification preferences, and investment decision support tools.
- **Tester/Energy engineer:** System maintenance, debugging and troubleshooting, performance optimization, database management, API integration, and technical documentation updates.

Table 2: Product Backlog table with its different actors and use stories

ID	Actor	User Story / Task	Priority
US-1	Investor	As a user, I want to create an account with email verification so I can securely access the platform.	Low
US-2	Investor	As a user, I want to view energy simulation data and manipulate it on my dashboard so I can analyze my results easily.	High
US-3	Investor	As a user, I want to improve loading performance of the imports/exports.	Medium
US-4	Admin	As an admin, I want to impersonate users for debugging purposes.	High
US-5	Tester	As a tester, I want to run automated tests on the energy simulation system to verify accuracy.	High
US-6	Energy Engineer	As an energy engineer, I want to simulate long-term usage to ensure endurance and stability.	High
US-7	System tester	As a system tester, I want to measure response time and energy calculation accuracy.	Medium

III. Conclusion

Sprint 0 establishes fundamental elements necessary for creating an efficient and productive working environment. The mission during this phase is to establish ground rules, integrate team members into the company's policies and standards, and configure the development infrastructure. For the entire project, we collaboratively defined the product backlog and determined all sprint schedules through team consensus. The subsequent chapters provide detailed discussions of individual sprints and their associated workloads.

Chapter 3: Conception

I. Introduction

In this chapter, we delve into the project more with specifying the workload and explaining the main functionalities done in the period of the 6-month internship. All the cases and diagrams of the project are related to unified modeling language (UML) [8], it helped during this final year project are explained in this chapter.

II. Maon web services

1. Sprints organization

As table 3 presents, during this work we fixed 5 sprints, each addressing multiple functionalities and each included in a module that categorizes the work structure to better understand the composure of the work done.

Table 3: Sprints organization table

Module	Sprint	Sprint Theme / Goal	Description / Focus
Module 1: Authentication	Sprint 1: Platform Accessibility	User Access & Security	Set up authentication system and focus on users' connectivity to ensure real and authentic traffic.
Module 2: Dashboard UI	Sprint 2: Dashboard & Data Enhancement	Front-End Dashboard Development	Modify and enhance the main dashboard interface. Integrate data visualization and CRUD components to display user and system data dynamically.
	Sprint 3: Fixes & Backend Management	Stabilization and Optimization	Fix UI/UX bugs, optimize dashboard performance, and connect Core application functionalities with the main web application with its different results.
Module 3: Backend & Admin Usage	Sprint 4: System Management & Admin Features	Back-Office and API Administration	Develop backend logic for admin control, implement data management endpoints, and create an admin dashboard for monitoring and managing system usage.

Module Energy Simulation Testing	4: Sprint Validation Performance Review	5: & Energy Simulation Application Testing	Test the Maon simulation application and ensure its credibility and its endurance with time and performance.
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2. Use case Diagram

The use case diagram offers a global understanding of the multiple functionalities used for this project. We'll find in the next parts the different actors, functionalities and relations between clients. For the global use case for the whole Maon system, we can find the Figure 5 that describes its functionalities and services provided to users as we will explain the use cases that we triggered in the work done for the internship and for the web application especially.

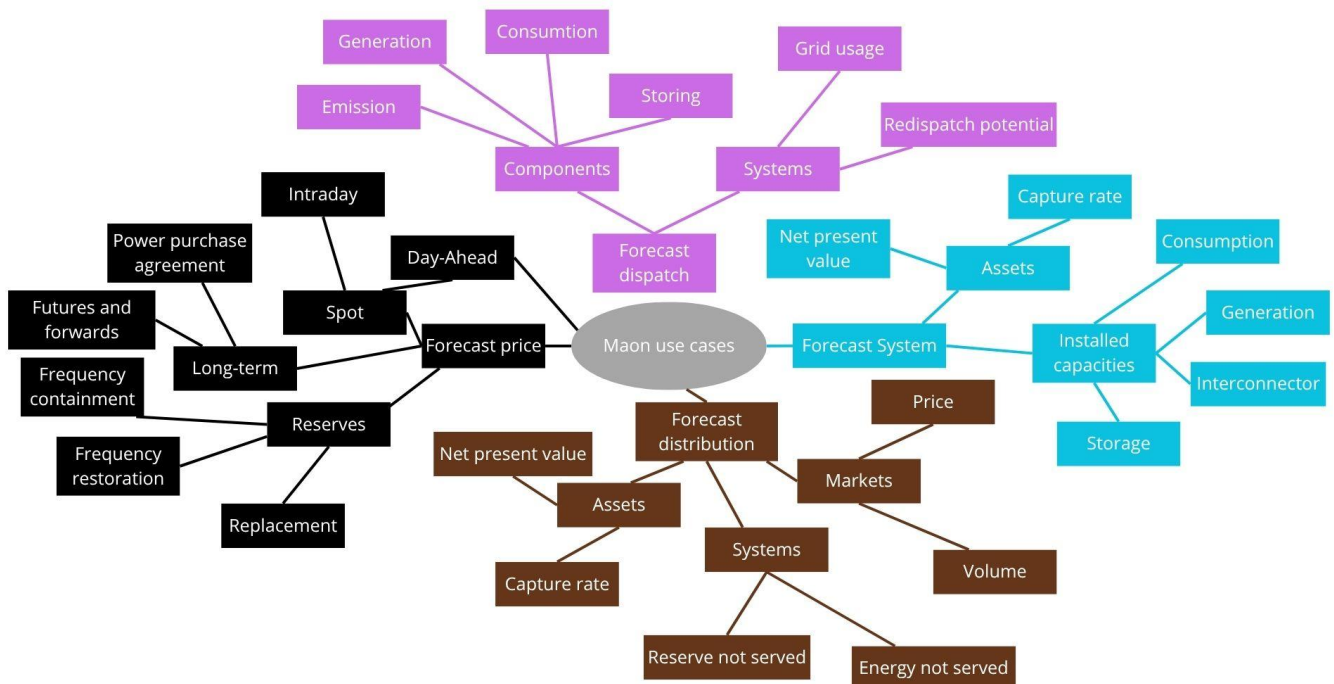


Figure 5: Maon's use cases diagram

a. Sprint 1: Platform Accessibility

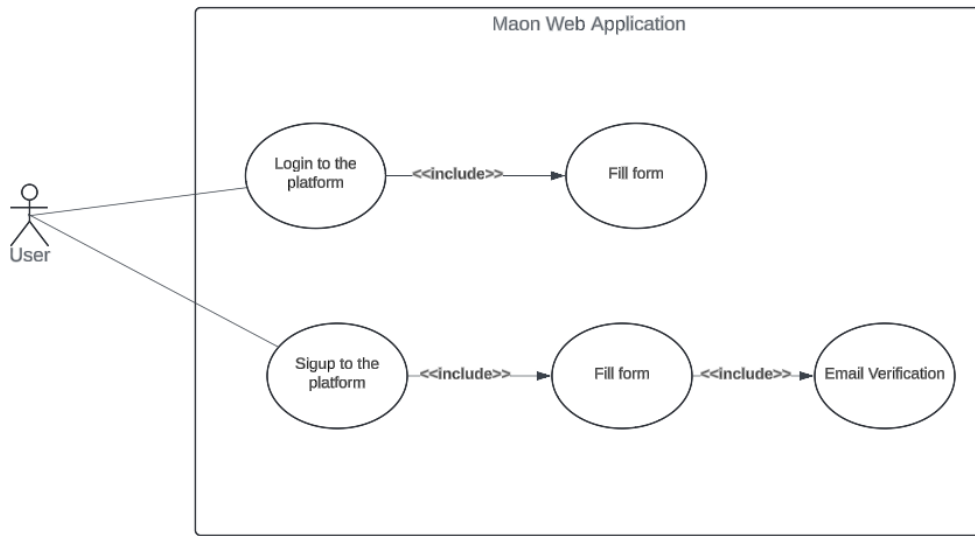


Figure 6: use case for the platform accessibility sprint

b. Sprint 2&3: Dashboard UI

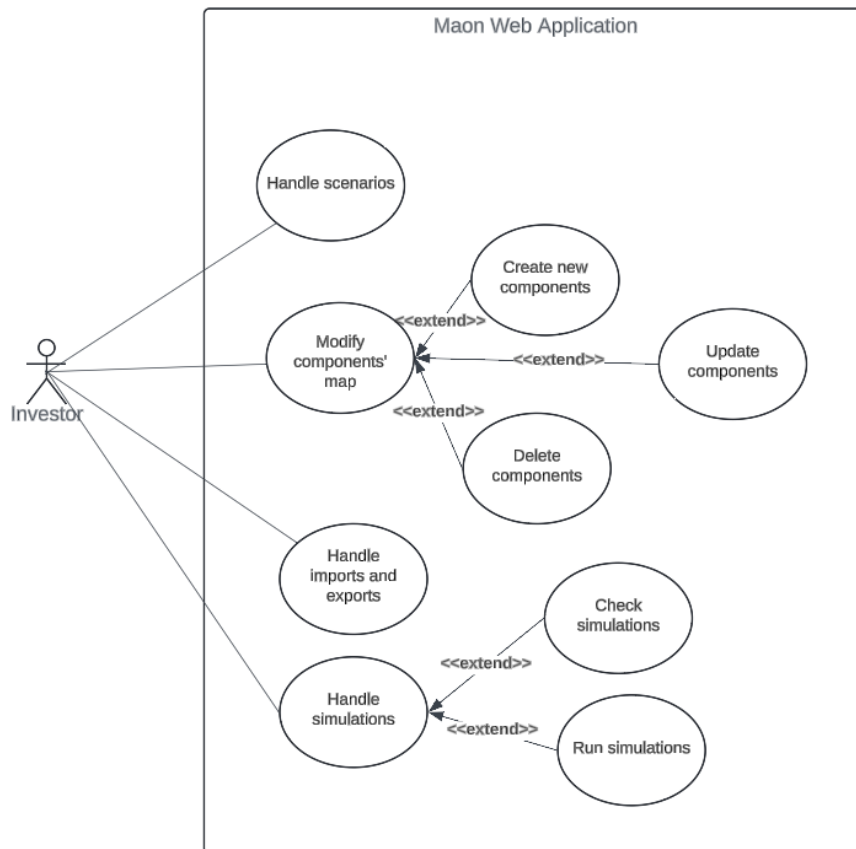


Figure 7: Use case for the Dashboard UI sprint

c. Sprint 4: System Management & Admin Features

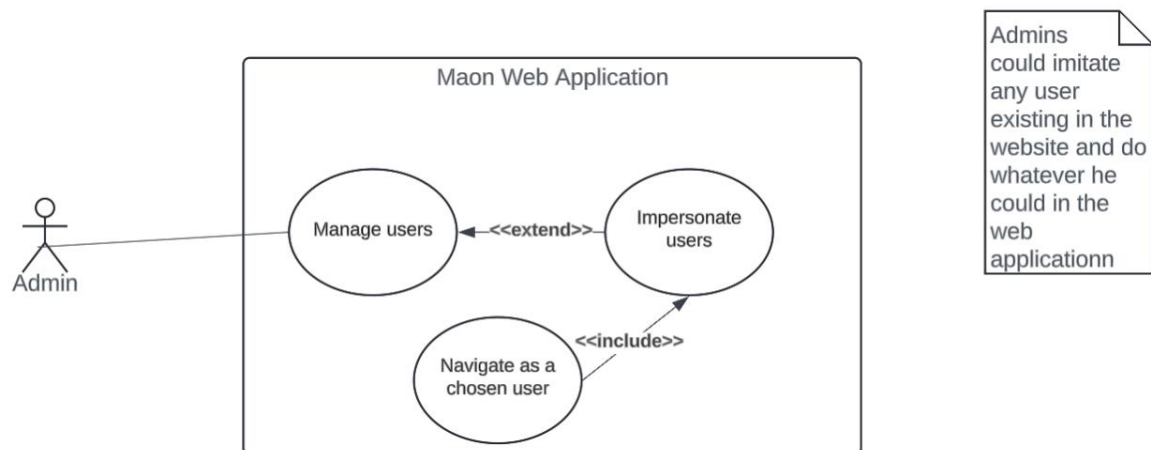


Figure 8: Use case for system management sprint

d. Sprint 5: Validation & Performance review

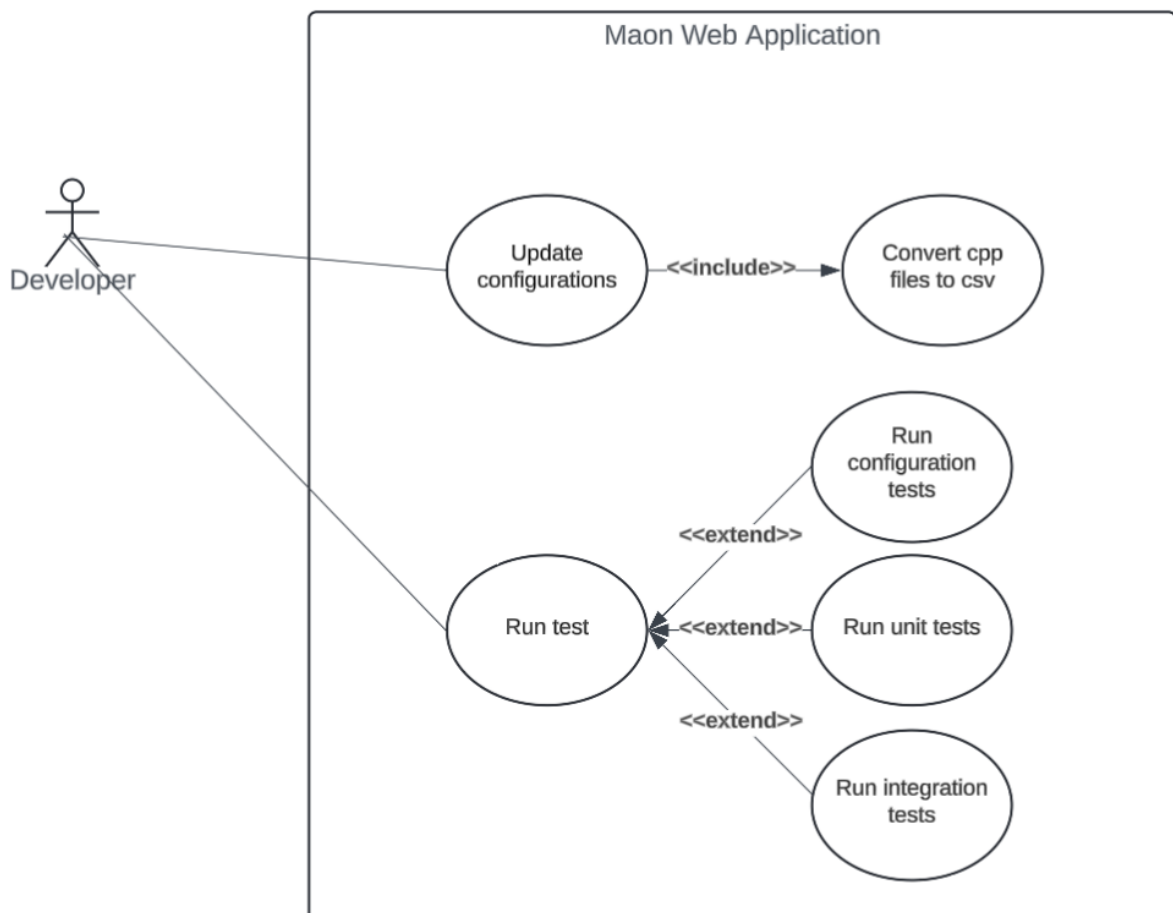


Figure 9: Use case for testing and performance review sprint

3. Activity Diagram

The Activity Diagram describes the multiple functionalities used in each Sprint. Here we can see scenarios that usually define the main services of the application and the interaction it can create to attain the objective needed. We're splitting the next parts to 2 describing the first and third sprints' different contributions to the project.

a. Sprint 1 : Platform Accessibility

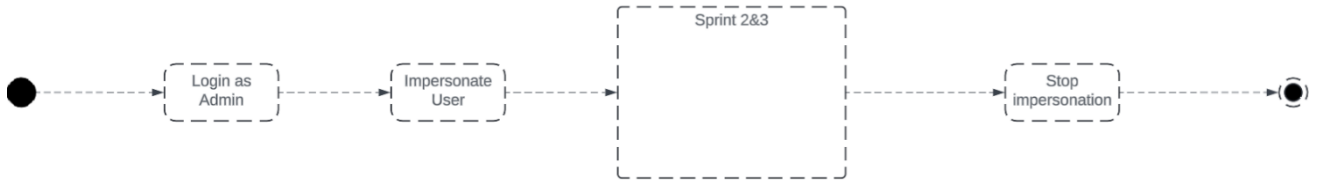


Figure 10: Activity diagram for platform accessibility

b. Sprint 5: Validation and performance review

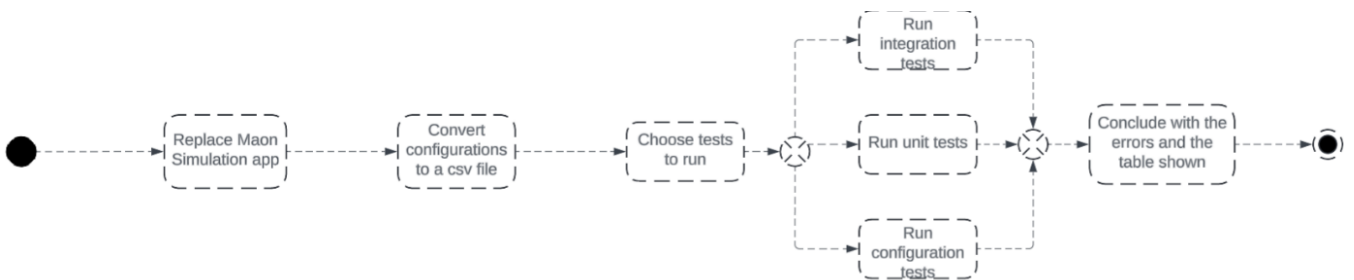


Figure 11: Activity diagram for testing and reviewing Core application

4. Class Diagram

A class diagram is a static, structural diagram within the Unified Modeling Language (UML) that illustrates a system's design by representing its classes, their attributes (data), operations (methods), and the relationships (associations, aggregations, and inheritance) between them.

Here we can see multiple classes:

- User: an individual account that can log in, access projects, run simulations, and view results.
- Dashboard: the first page after login, giving an overview of the projects, simulations, and storage use.
- Setting: configuration interfaces (for user, cloud, or project scope) that let users adjust preferences (e.g. UI, formats) and manage version control
- Admin: a role or user with elevated permissions (impersonating and managing users, projects or global settings).
- Project: a container grouping scenarios and configurations; it is the root for scenario trees.

- Simulation: a run or snapshot of a scenario's inputs and outputs, executed to produce results.
- Scenario: a set of model components and simulation runs within a project; can inherit from parent scenarios.
- Tester: typically someone who verifies correctness or quality of simulations run in the platform.

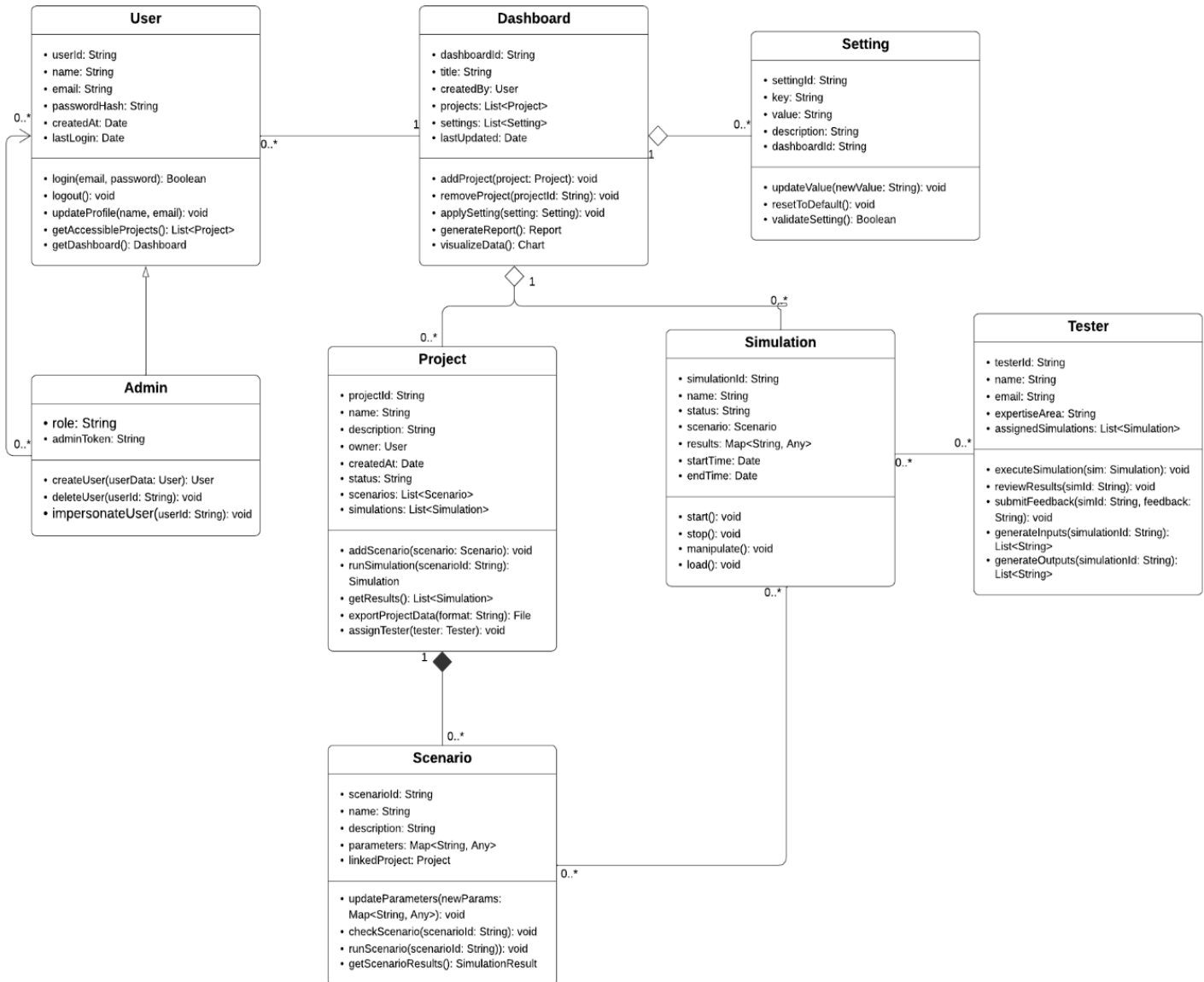


Figure 12: Global class diagram for Maon web application

5. Sequence Diagram

The sequence diagram illustrates the dynamic interactions between the system's core components over time. It focuses on how users, admins, and system modules communicate to achieve key functionalities within the platform. In this context, the primary emphasis is placed on platform accessibility and the account confirmation process, which ensure secure and validated user access to the system.

Additionally, a second diagram highlights workflows from Sprint 4 – Admin Management, where the admin role gains advanced control features, including user impersonation. This functionality allows administrators to temporarily access and manage user accounts for support, troubleshooting, and system validation purposes, ensuring efficient platform maintenance and enhanced user experience.

e. Sprint 1: Platform Accessibility

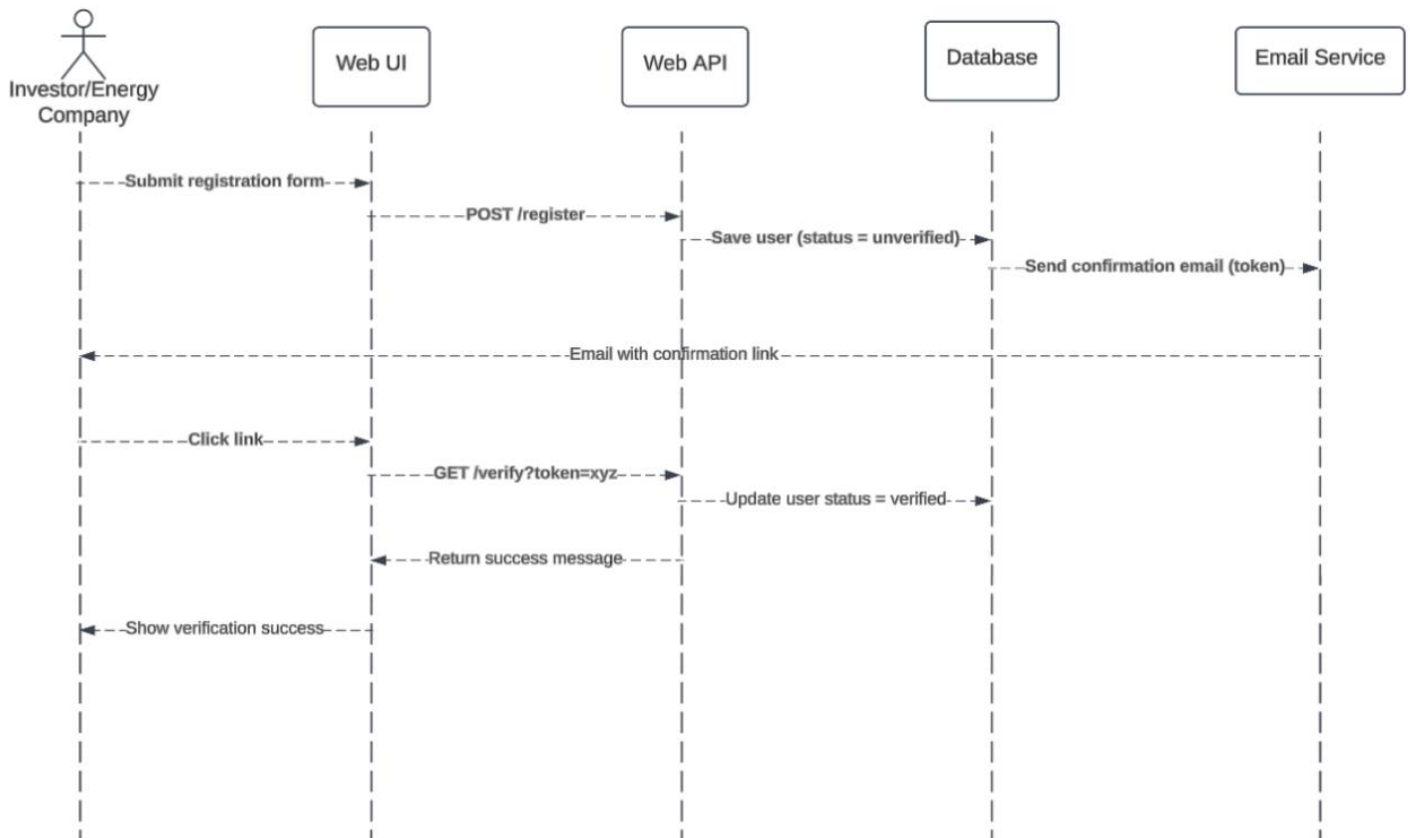


Figure 13: Sequence Diagram for registering into the company's platform

f. Sprint 4: System Management & Admin Features

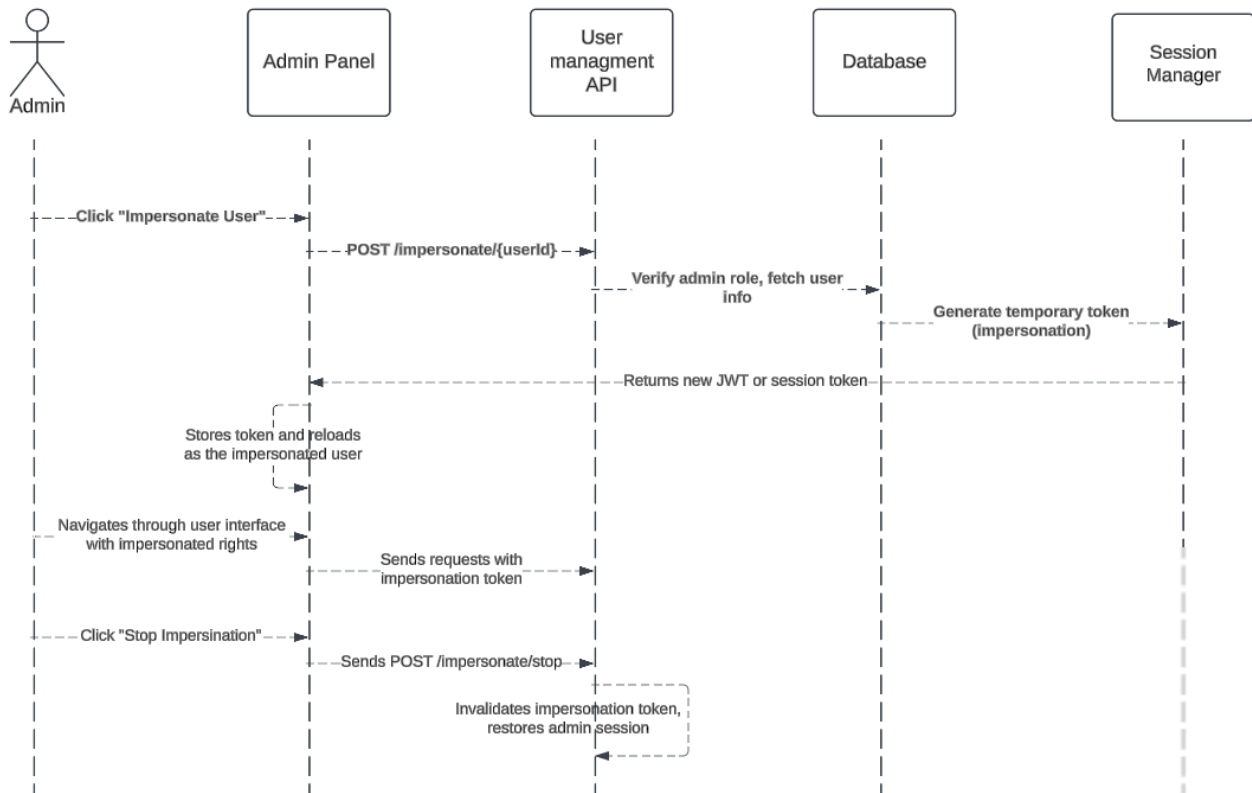


Figure 14: Sequence diagram for admin feature (impersonating users)

III. Conclusion

In the conception chapter, we encountered a lot of functionalities provided by Maon Web Application to understand better its uses and different interactions with users. Different diagrams were used such as use cases, activity diagrams, class diagrams and sequence diagrams. Each one has its own integration in the web app and by providing them, the project would seem a lot easier to realize. Which would be the next chapter for this report.

Chapitre 4: Work achieved

I. Introduction

The last chapter consists of displaying the realization of the web application and how our involvement and contribution in it helped to enhance it from bugs and previous work.

In this chapter we'll explore the working environment and different technologies used with figures of some tasks held during the period of the internship.

II. Material Environment

This project was used on multiple laptops connected to Maon's server:

Table 4: Maon's laptop criteria

Processeur	ASUS Laptop Zenbook A14 UX3407QA
RAM	16 Go
Disque Dur	1 To
Système d'exploitation	Arch Linux (Manjaroo)

III. Environment logiciel

To understand better the functions of the web application, we used the following technologies and frameworks.

1. Frameworks

The developers worked with MERN stack to provide this web application; we can see the different technologies from this stack:

- **MongoDB:** NoSQL database used to store MAON's project data in a flexible and scalable way.
- **Express.js:** Backend framework running on Node.js, handling MAON's API logic and server-side routing.
- **ReactJS:** Frontend library used to build the company's interactive user interfaces and dashboards.

- **Node.js:** JavaScript runtime environment used to execute server-side code efficiently.

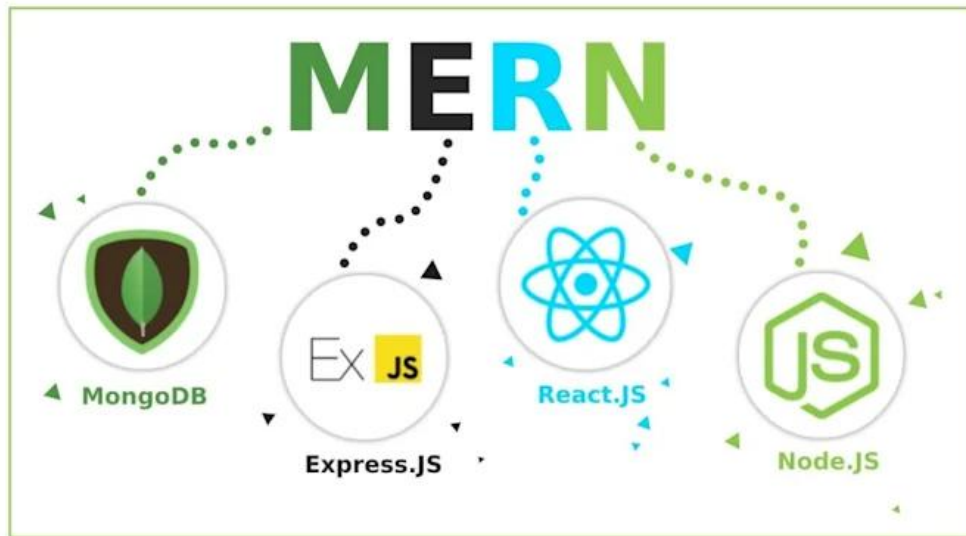


Figure 15: MERN stack logo

2. Outils Technique

- **VsCode:** The main development environment for writing, debugging, and maintaining MAON's codebase.

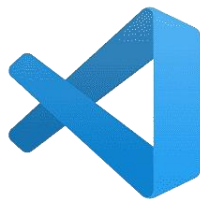


Figure 16: VSCode logo

- **Mongo Compass:** the graphical user interface tool is used to interact with MongoDB databases

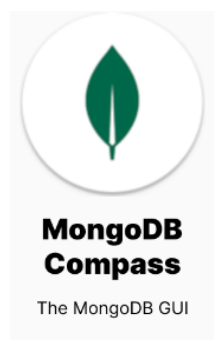


Figure 17: MongoCompass logo

- **Telepresence [9]:** Tool used in MAON for testing and debugging Kubernetes services directly from the local development environment.



Figure 18: Telepresence logo

- **Apollo Server [10]:** Middleware that integrates GraphQL into MAON's backend to manage data queries and mutations.



Figure 19: Apollo server logo

- **GraphQL:** Query language that enables efficient communication between the company's frontend and backend.



Figure 20: GraphQL logo

- **MiniIO [11]:** Object storage system used in MAON for managing simulation files and large datasets.



Figure 21: Minio logo

- **BootstrapCSS [12]:** Framework used to design responsive and consistent UI components for MAON's web interface.



Figure 22: Bootstrap logo

-
- **GitLab [13]:** Platform for source code management, CI/CD pipelines, and collaboration within the MAON development team.



Figure 23: Gitlab logo

- **Git:** Version control system used to track and manage changes across MAON's codebase.



Figure 24: Git logo

3. Research and Analysis Tools

- **Chatgpt:** AI assistant created by the company OpenAI [14], used to support documentation writing, code refactoring, and idea generation during MAON's development.



Figure 25: Chatgpt logo

- **Claude.ai:** AI tool used alongside ChatGPT for reviewing documentation and generating technical content for MAON.



Figure 26: Claude logo

- **Maon Documentation [15]:** Official reference portal providing guidelines, architecture overviews, and usage examples for the MAON platform.

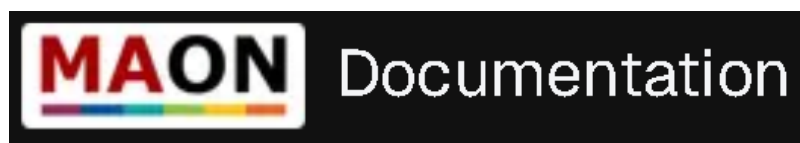


Figure 27: Maon Documentation logo

IV. Realization

For this last chapter, we'll take a closer look at the platform and the different functionalities held by me during the internship. The work shown is from the main version of the application where some of the functionalities done are better explained.

1. Gitlab progress

Maon GmbH used GitLab as a development environment to handle issues, tickets and bugs and approve merge requests and review them. Below is my workload for this period, with each day color-coded (the darkest the color, the higher the contribution) to represent the level of contributions made:

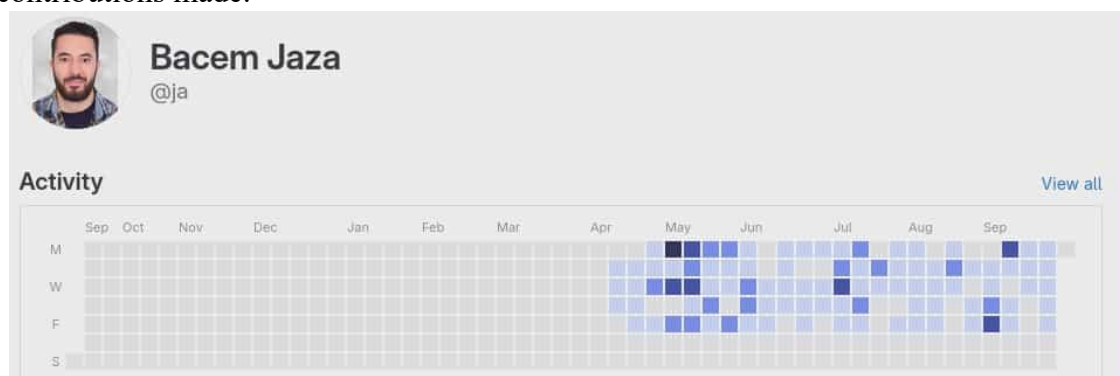


Figure 28: Contribution in Gitlab

I've maintained and fixed a lot of issues for this project and merged them afterwards into the main web application with the approval of Maon's head developer; we can mention some of them next:

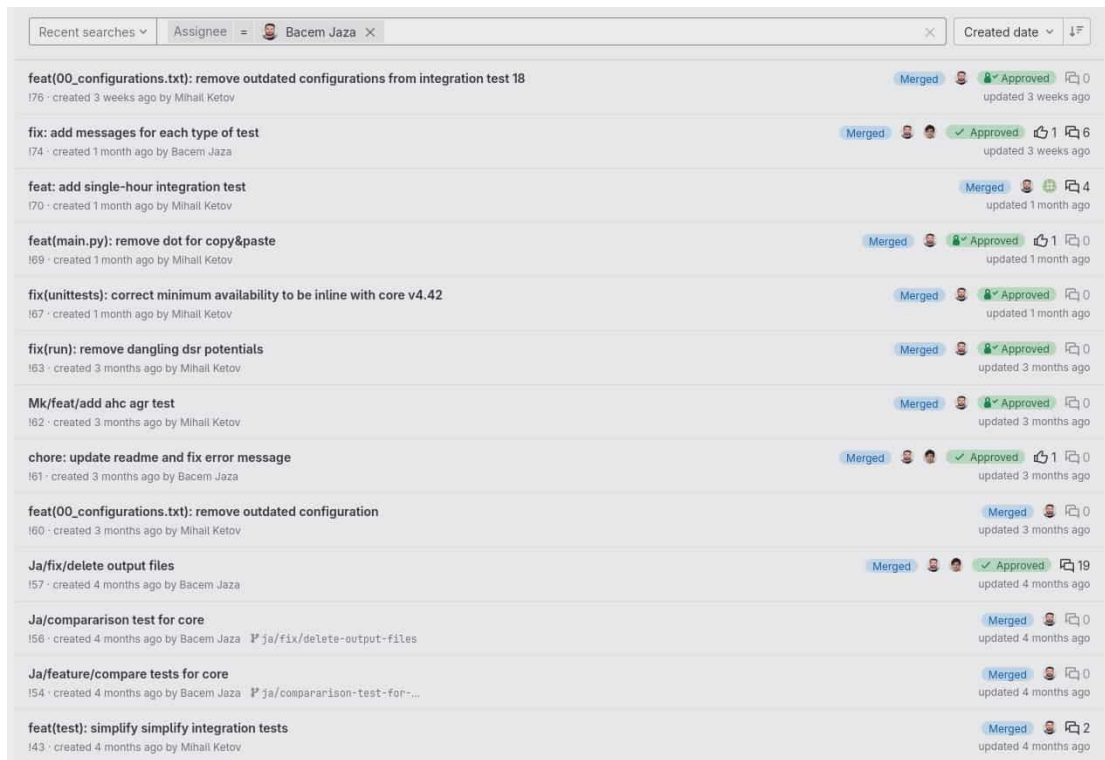


Figure 29: Merge requests for frontend application

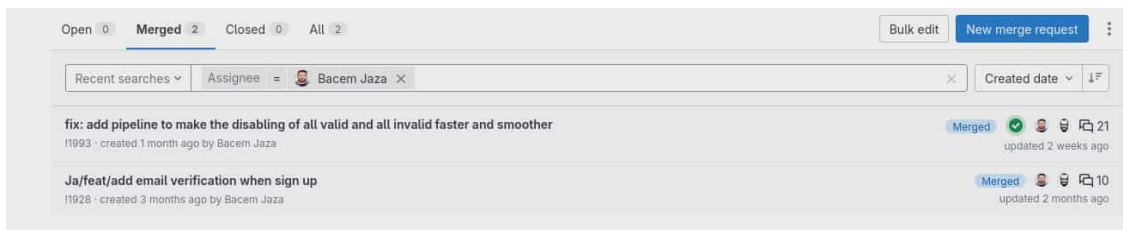


Figure 30: Merge requests for backend application

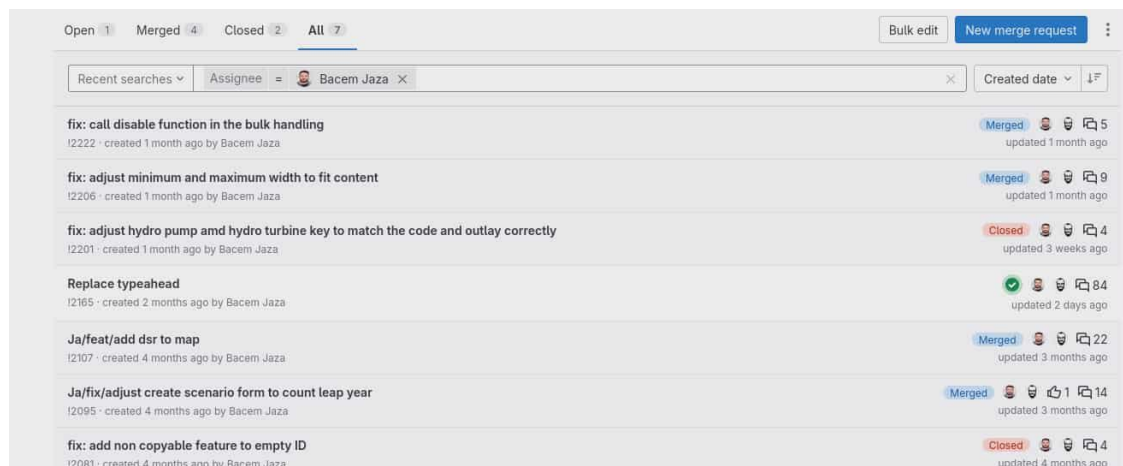


Figure 31: Merge requests for testing application

2. Data structure

The overall structure of the data flow at the web-based platform Maon can be seen in the following figure. It displays the different layers that need to be done by the application to get the correct outputs for the web application.

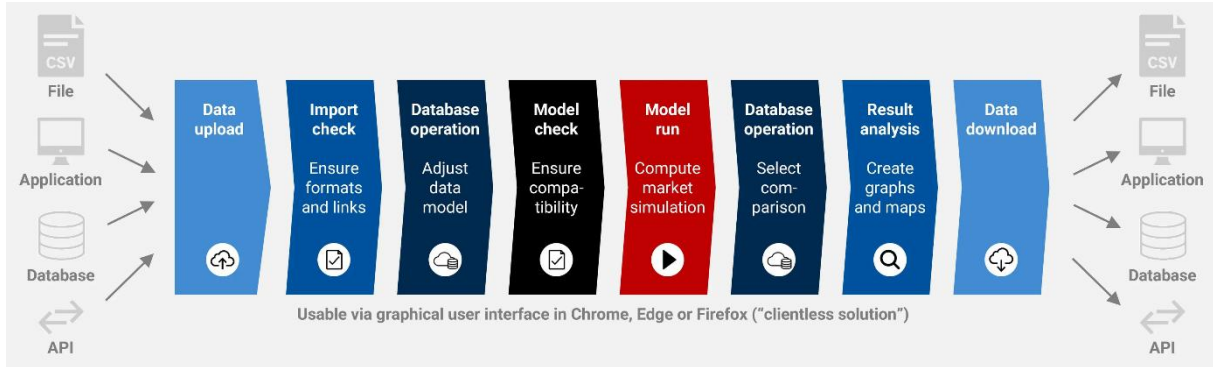


Figure 32: Maon application Data structure

3. Technologies used for Maon

To show the work done, it's a must that we introduce the different technologies and components that the platform is based on. These are crucial elements that determine the platform's efficiency, and they represent as follows:

Each of the elements produces outputs that are effective for the process of the web application.

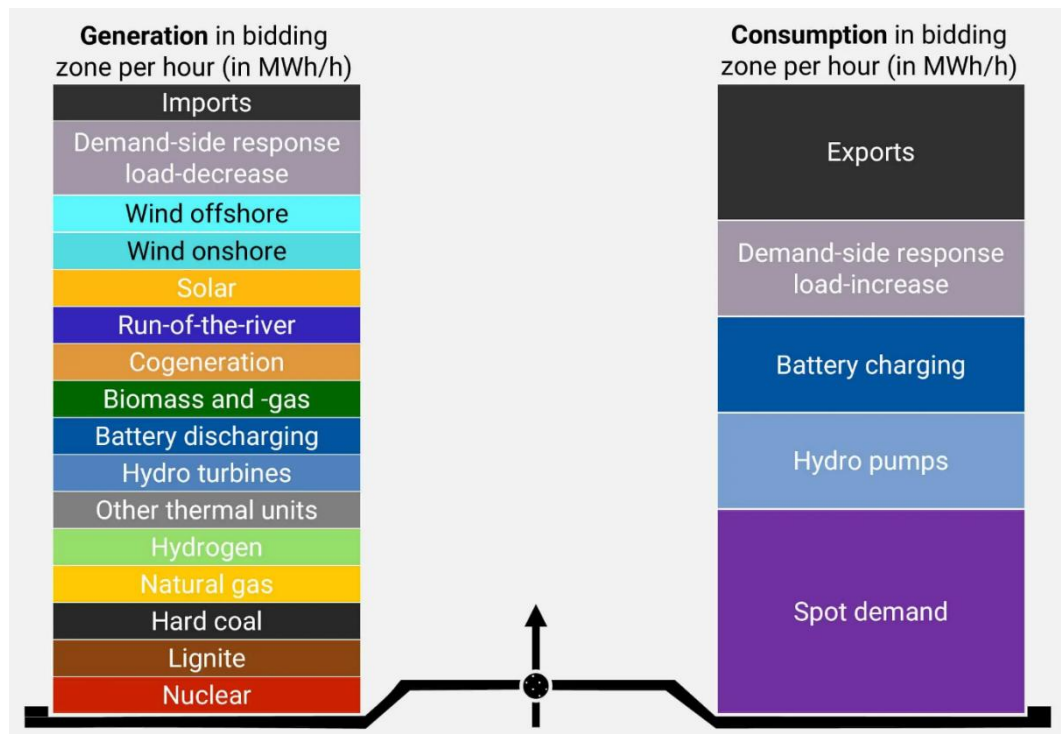


Figure 33: Example of energy balance with various generation and consumption technologies

4. Work achieved

Here, we'll show some of the work done for each module, we focused on some of the most effective and important tasks held and done during this internship with integrating in the final project. These tasks are:

Module 1:

- Email Verification and form correction when signing up

Users for Maon's platform could now receive an email verification when signing up, to confirm the user's email. Also, it makes Maon's developers sure of the registered users information and avoid non active users.

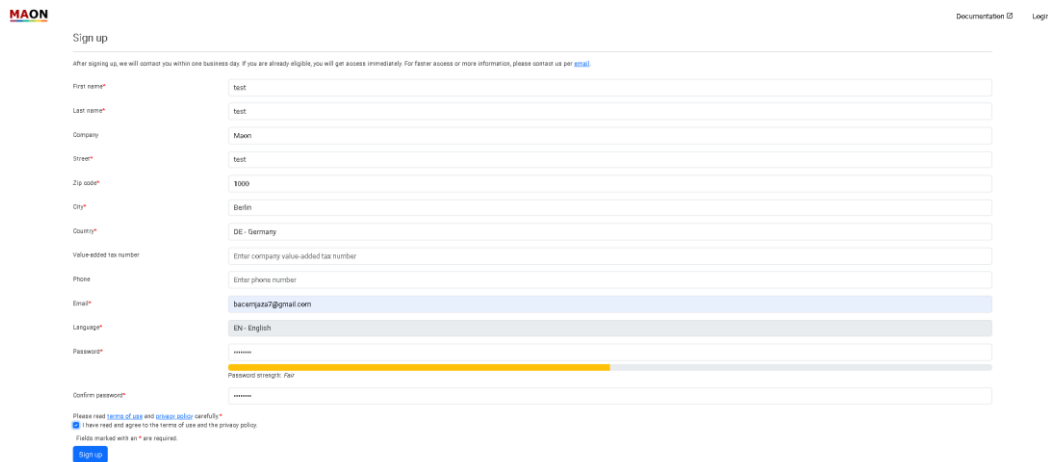
A screenshot of the Maon web application's sign-up form. The form is titled 'Sign up' and includes a sub-header: 'After signing up, we will contact you within one business day if you are already eligible, you will get access immediately. For feature access or more information, please contact us per [email](#).' The form fields are: First name* (test), Last name* (test), Company (Maon), Street* (test), Zip code* (1000), City* (Berlin), Country* (DE - Germany), Value-added tax number (Enter company value-added tax number), Phone (Enter phone number), Email* (baccapata7@gmail.com), Language* (EN - English), Password* (with a strength indicator bar), and Confirm password*. At the bottom, there is a checkbox for 'I have read and agree to the terms of use and the privacy policy' and a 'Sign up' button.

Figure 34: Account registration for Maon's web application



We've sent you an email verification, please click the link sent with the email in order to verify your email address.

Figure 35: Frontend display of a sent message in the email registered for the user

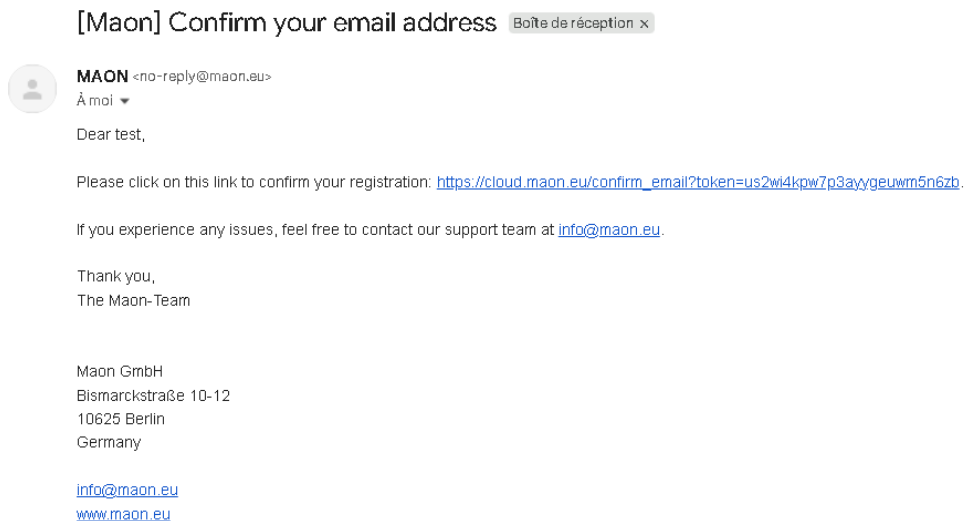


Figure 36: Email confirmation for registration

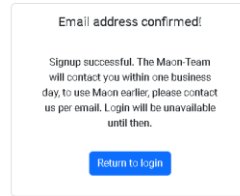


Figure 37: Account confirmation after clicking email link

Module 2:

- Bulk Handling:

This bulk handling issue consists of managing multiple components existing in a particular scenario. The program is supposed to disable all, disable all derived, disable all valid and all invalid components, also enabling and deleting all of them. The validity of the component consists of the scenario year where it's valid if it's in the zone of commission and decommission year. The derived components means that they did change from the parent scenario and got their own customization for this year.

MAON Documentation Dashboard John Doe

>> 2032 - rth Input Simulation Output

Create Delete Disable

Edit battery storages

Column visibility Show filters Reset view

☐			Bidding Zone	Battery	Tech	Capacity (MWh)	P Max Charge (MW)	
☐	↓	✓	AT	qq	RF (Redox-Flow)	5000	500	
☐	↑	✓	AT	dd	LI (Lithium-Ion)	800	8000	
☐	↓	✓	AT	gg	SS	2000	20000	

10 0 / 3 Rows Selected 1 of 1

Figure 38: Components display

MAON Documentation Dashboard John Doe

>> 2032 - rth Input Simulation Output

Bulk handling of components

☐	File	Delete	Disable	Enable	Valid	+	Invalid	+	Disabled	=	Origin	+	Derived	=	Total
☐	10_demands_spot.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	11_demands_fcr.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	12_demands_afrr_positive.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	13_demands_afrr_negative.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	14_demands_mfrr_positive.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	15_demands_mfrr_negative.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	20_dsr_consumers.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	21_dsr_potentials.csv	Delete	Disable all	Enable	0		0		0		0		0		0
☐	22_dsr_mustruns_outages_revisions.csv	Delete	Disable all valid	Enable	0		0		0		0		0		0
☐	23_dsr_restrictions_work.csv	Delete	Disable all invalid	Enable	0		0		0		0		0		0
☐	24_dsr_availabilities.csv	Delete	Disable all origin	Enable	0		0		0		0		0		0
☐	30_battery_storages.csv	Delete	Disable	Enable	3		0		0		1		2		3
☐	31_battery_mustruns_outages_revisions.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	32_battery_states_of_charge.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐	33_battery_availabilities.csv	Delete	Disable	Enable	0		0		0		0		0		0

Figure 39: Bulk Handling of components page

MAON

Documentation Dashboard John Doe

2032 - rth
Input Simulation Output

Bulk handling of components

☐ File	Delete	Disable	Enable	Valid	+	Invalid	+	Disabled	=	Origin	+	Derived	=	Total
☐ 10_demands_spot.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 11_demands_fcr.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 12_demands_afrr_positive.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 13_demands_afrr_negative.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 14_demands_mfrr_positive.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 15_demands_mfrr_negative.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 20_dsr_consumers.csv	Delete	Disable	Enable	0		0		0		0		0		0
☐ 21_dsr_potentials.csv	Delete	Disable all	Enable	0		0		0		0		0		0
☐ 22_dsr_mustruns_outages_revisions.csv	Delete	Disable all valid	Enable	0		0		0		0		0		0
☐ 23_dsr_restrictions_work.csv	Delete	Disable all invalid	Enable	0		0		0		0		0		0
☐ 24_dsr_availabilities.csv	Delete	Disable all origin	Enable	0		0		0		0		0		0
☐ 30_battery_storages.csv	Delete	Disable all derived	Enable	3		0		0		1		2		3

Figure 40: Result after disabling all derived components

☐ 30_battery_storages.csv

Delete

Disable

Enable

1

0

2

3

0

3

Figure 41: Disable all derived for battery storage component

- Enhancing the upload of output compare files:

When uploading output compare files, the platform used to freeze, and no loading screen is shown. Also, the only possible way is to refresh the page to see updates. You'll find below the updated version of the upload, dealing with data with much more efficiency.

MAON

Documentation Dashboard John Doe

2036 - wef
Input Simulation Output

Output compare files

Upload

Column visibility Show filters Reset view

File name	Description	Type
No data to show		

10 0 Rows
1 of 0

Create output compare file

Description

description

Type*

Assessment thermal power pla

File*

Browse... 85_therm_sion.csv

File Name

85_thermal_restrictions_emission.x

Fields marked with an * are required.

Save

Figure 42: Upload output compare files

MAON

Documentation Dashboard John Doe

2036 - wef
Input Simulation Output

Output compare files

Upload

Column visibility Show filters Reset view

File name	Description	Type
Loading data...		

10 0 Rows
1 of 0

Figure 43: Loading screen when uploading of output compare file

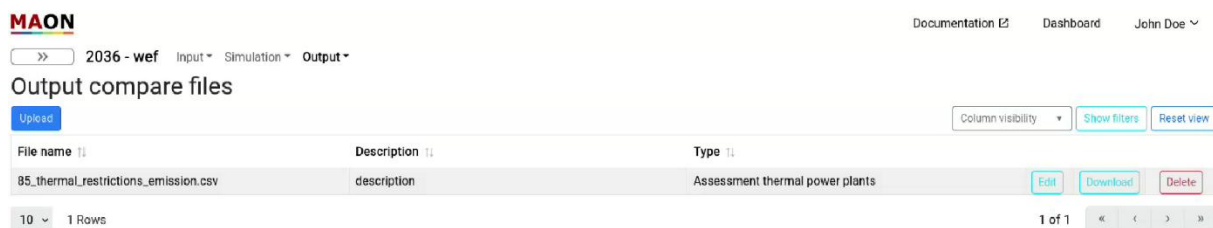


Figure 44: Display of uploaded output compare file

- Handle scenario years:

Scenario years inheritance was a bit buggy at first, it didn't count leap years and wasn't UI/UX oriented. With the new form to create scenarios, now it seems very easy and adaptable to clients' desires with a much more dynamic years counting when picking parent.

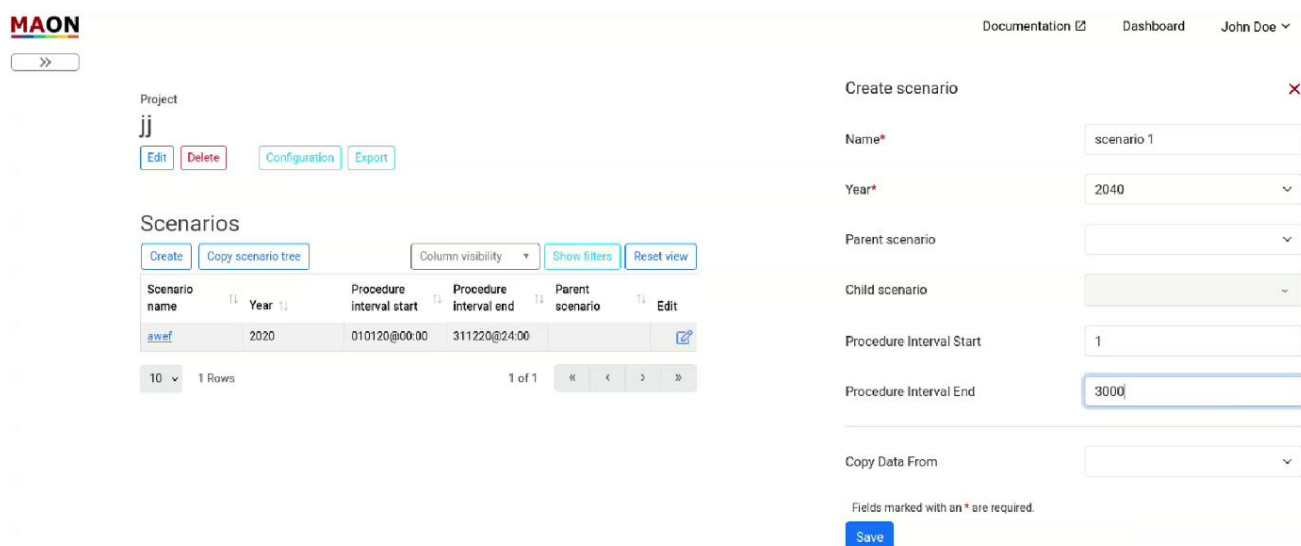


Figure 45: Create a new scenario

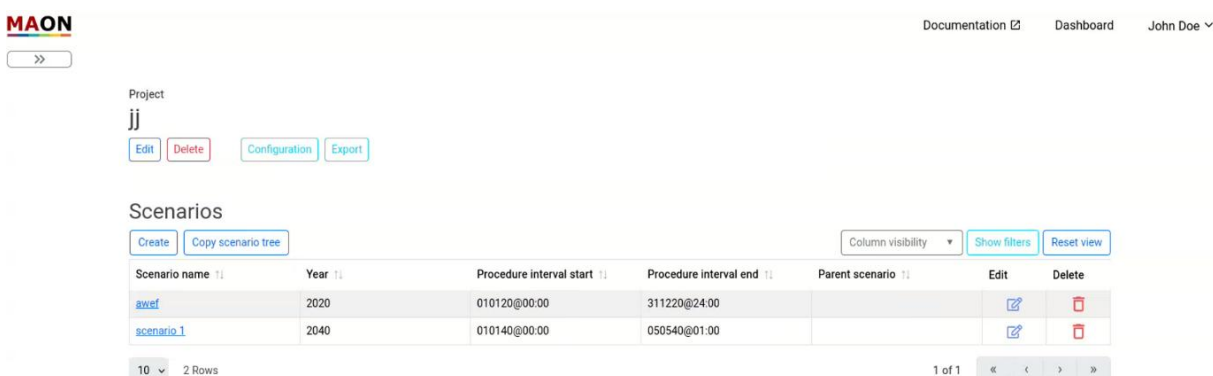


Figure 46: Scenario created successfully

Project: jj

Scenarios

Scenario name	Year	Procedure interval start	Procedure interval end	Parent scenario	Edit
awef	2020	010120@00:00	311220@24:00		Edit
scenario 1	2040	010140@00:00	050540@01:00		Edit

10 2 Rows 1 of 1

Create scenario

Name* scenario 2

Year* 2061

Parent scenario 2040 - scenario 1

Child scenario

Procedure Interval Start 010161@00:00 or 1

Procedure Interval End 311261@24:00 or 8760

Copy Data From

Fields marked with an * are required.

Save

Figure 47: Create a new scenarion while selecting the previous one as parent

Project: jj

Scenarios

Scenario name	Year	Procedure interval start	Procedure interval end	Parent scenario	Edit
awef	2020	010120@00:00	311220@24:00		Edit
scenario 1	2040	010140@00:00	050540@01:00		Edit

10 2 Rows 1 of 1

Create scenario

Name* scenario 2

Year* 2061

Parent scenario 2040 - scenario 1

Child scenario

Procedure Interval Start 010161@00:00

Procedure Interval End 060561@01:00

Copy Data From

Fields marked with an * are required.

Save

Figure 48: Auto complete for the procedure interval start and end based on the parent's attributes

Project: jj

Scenarios

Scenario name	Year	Procedure interval start	Procedure interval end	Parent scenario	Edit	Delete
awef	2020	010120@00:00	311220@24:00		Edit	Delete
scenario 1	2040	010140@00:00	050540@01:00		Edit	Delete
scenario 2	2061	010161@00:00	060561@01:00	scenario 1	Edit	Delete

10 3 Rows 1 of 1

Figure 49: Scenario created successfully with respecting parent's details

- Fixing forms:

My role here consisted of making the forms smoother and efficient, especially in inputs that have multiple selections. The code that was used here was reliant on Typeahead library that used integrated functions that were buggy and not the desired solution.

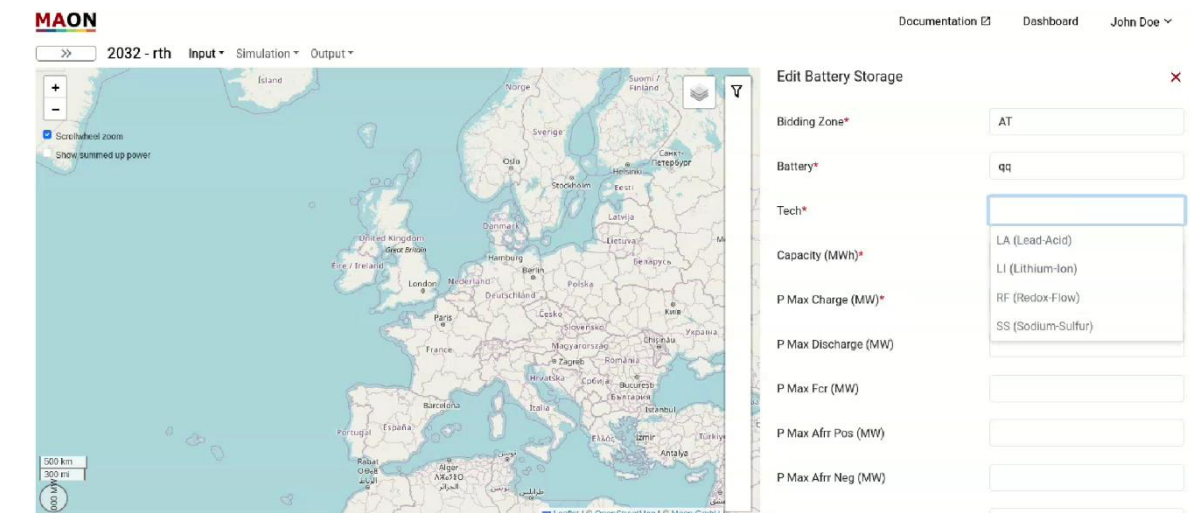


Figure 50: Focus on Tech input

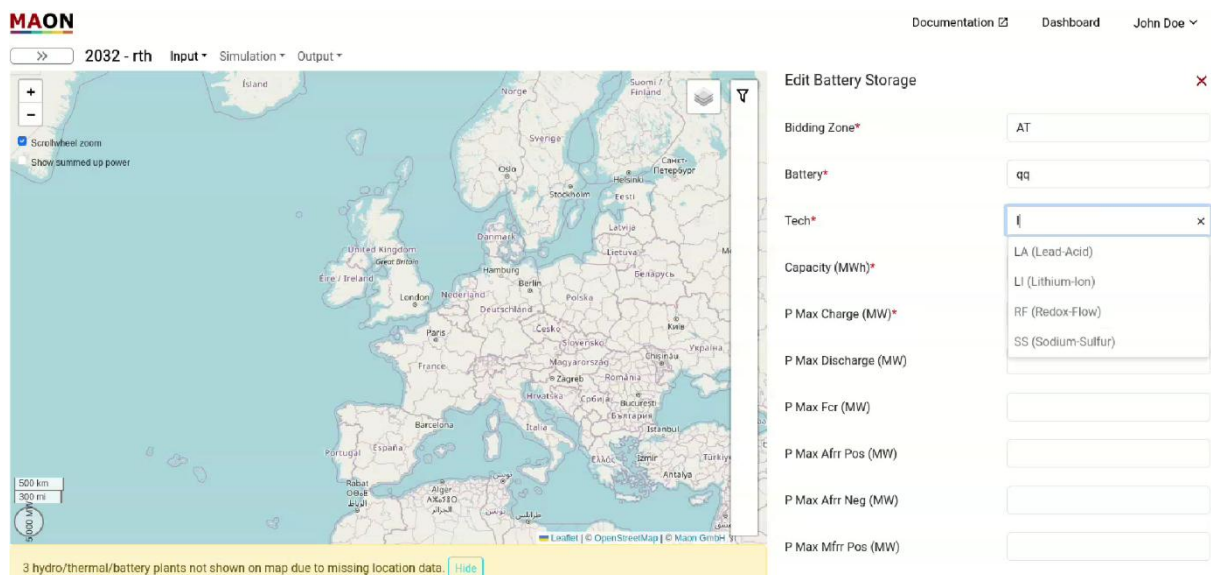


Figure 51: Write desired tech input and see the search done for the options

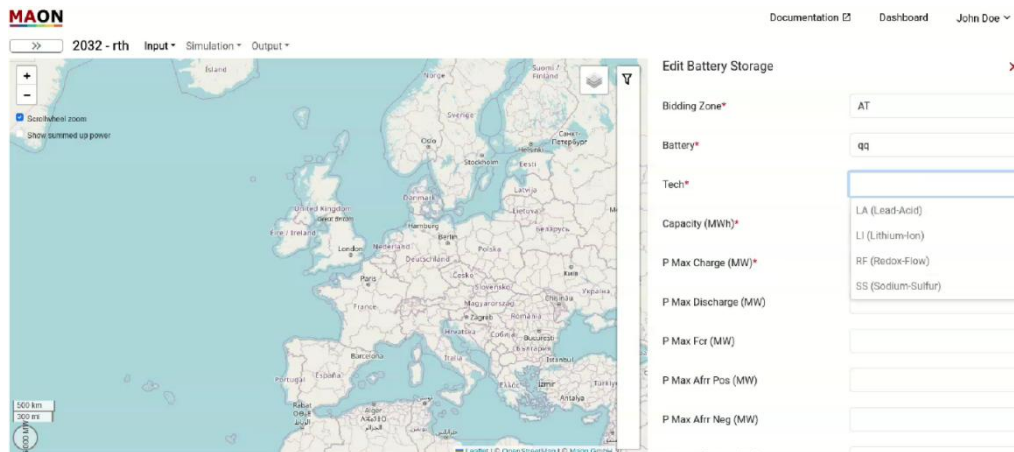


Figure 52: Delete input and navigate through the options with up and down buttons

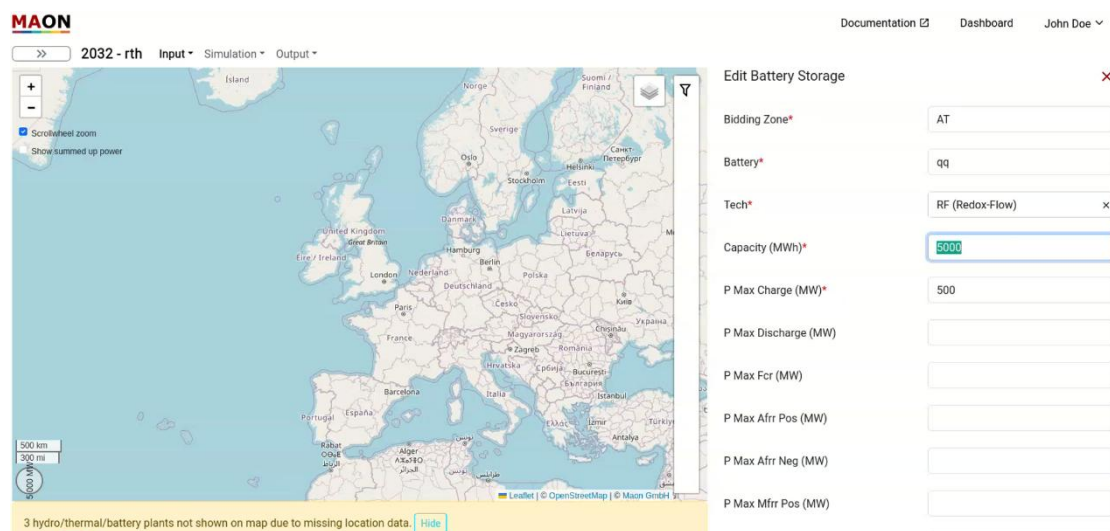


Figure 53: Click 'tab' button and focus on the next input

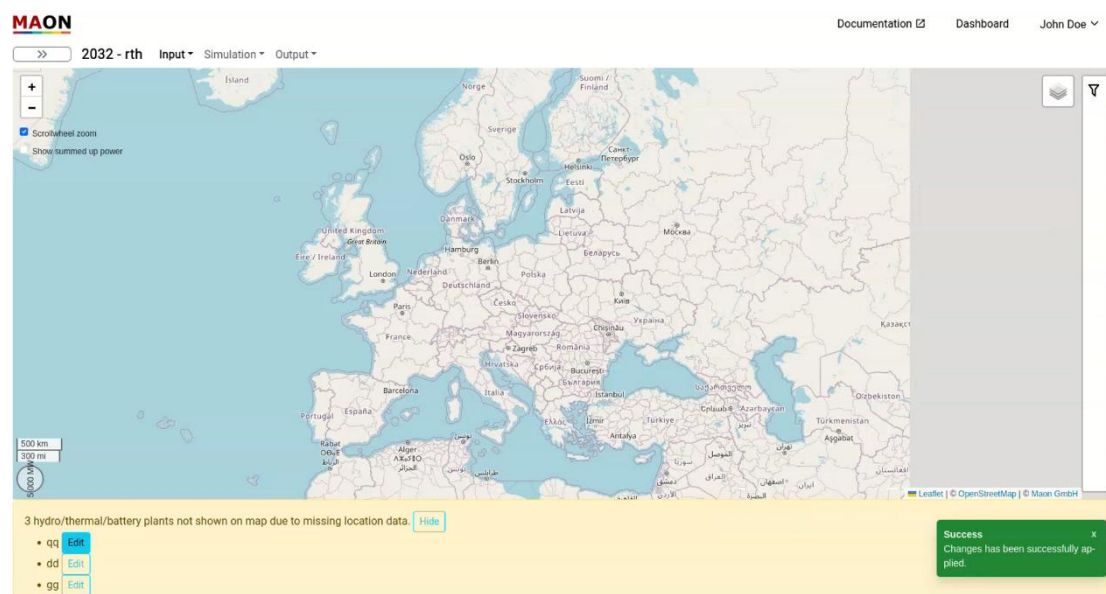
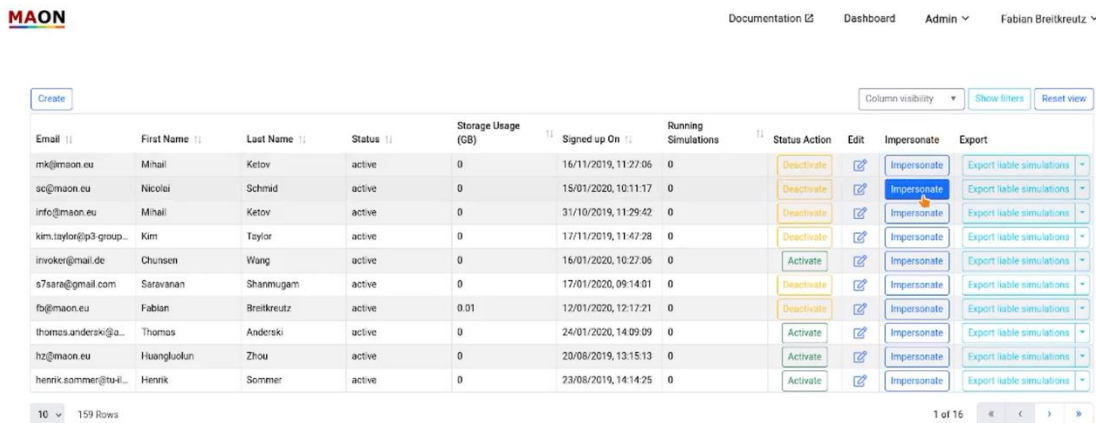


Figure 54: Press enter and successfully modify the tech attribute

Module 3:

- Impersonate User

Impersonating users is a crucial function for the admin that allows him to access every user's database and functionalities to correct, troubleshoot or understand better the client's dashboard. This makes the work easier and efficient for both the client and admins.



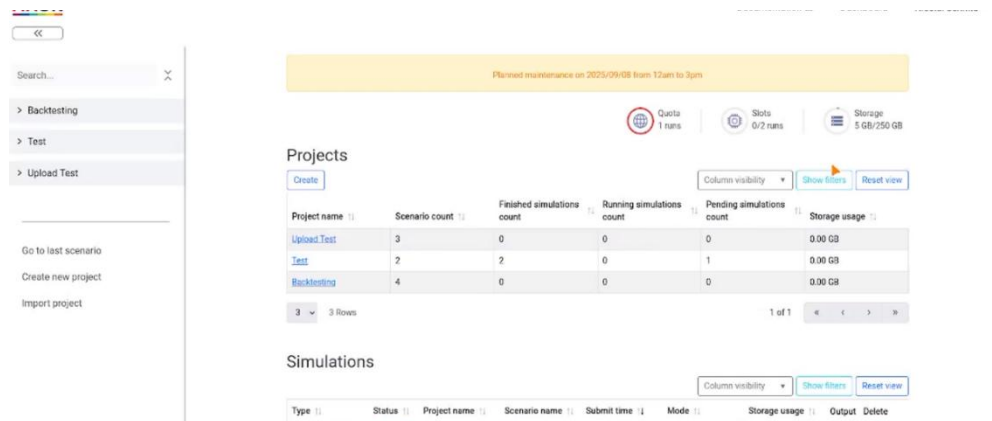
MAON Documentation Dashboard Admin Fabian Breikreutz

Create Column visibility Show filters Reset view

Email	First Name	Last Name	Status	Storage Usage (GB)	Signed up On	Running Simulations	Status Action	Edit	Impersonate	Export
mk@maon.eu	Mihail	Ketov	active	0	16/11/2019, 11:27:05	0	Deactivate		Impersonate	Export liable simulations
sc@maon.eu	Nicolai	Schmid	active	0	15/01/2020, 10:11:17	0	Deactivate		Impersonate	Export liable simulations
info@maon.eu	Mihail	Ketov	active	0	31/10/2019, 11:29:42	0	Deactivate		Impersonate	Export liable simulations
kim.taylor@p3-group...	Kim	Taylor	active	0	17/11/2019, 11:47:28	0	Deactivate		Impersonate	Export liable simulations
invoker@mail.de	Chunson	Wang	active	0	16/01/2020, 09:27:05	0	Activate		Impersonate	Export liable simulations
s7sara@gmail.com	Saravanan	Shanmugam	active	0	17/01/2020, 09:14:01	0	Deactivate		Impersonate	Export liable simulations
fb@maon.eu	Fabian	Breikreutz	active	0.01	12/01/2020, 12:17:21	0	Deactivate		Impersonate	Export liable simulations
thomas.anderski@a...	Thomas	Anderski	active	0	24/01/2020, 14:09:09	0	Activate		Impersonate	Export liable simulations
hz@maon.eu	Huangluokun	Zhou	active	0	20/08/2019, 13:15:13	0	Activate		Impersonate	Export liable simulations
henrik.sommer@tq4...	Henrik	Sommer	active	0	23/08/2019, 14:14:25	0	Activate		Impersonate	Export liable simulations

10 159 Rows 1 of 16

Figure 55: Navigate to admin panel and click on impersonate selected user



MAON Search... Backtesting Test Upload Test Go to last scenario Create new project Import project

Planned maintenance on 2025/09/08 from 12am to 3pm

Quota 1 nms Slots 0/2 nms Storage 5 GB/250 GB

Projects Create Column visibility Show filters Reset view

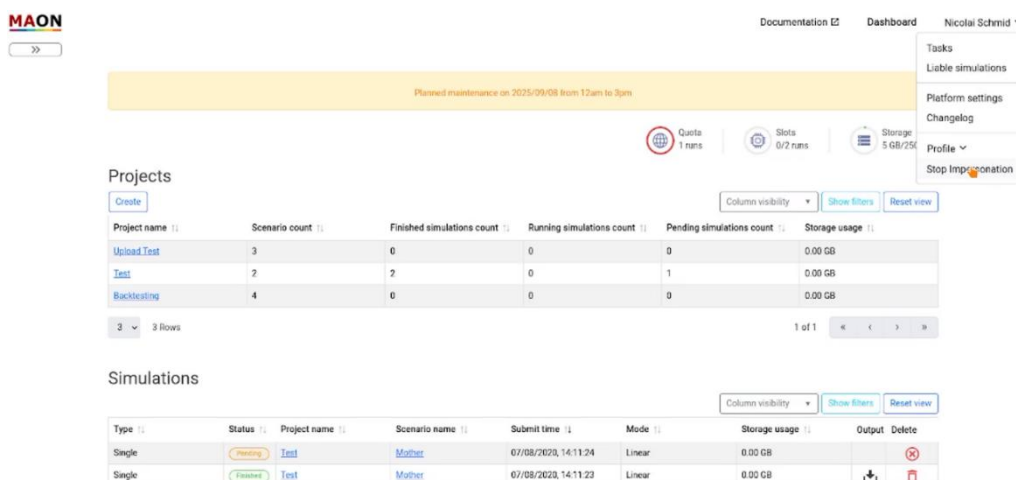
Project name	Scenario count	Finished simulations count	Running simulations count	Pending simulations count	Storage usage
Upload Test	3	0	0	0	0.00 GB
Test	2	2	0	1	0.00 GB
Backtesting	4	0	0	0	0.00 GB

3 3 Rows 1 of 1

Simulations Column visibility Show filters Reset view

Type	Status	Project name	Scenario name	Submit time	Mode	Storage usage	Output	Delete
Single	Running	Test	Mother	07/08/2020, 14:11:24	Linear	0.00 GB		
Single	Failed	Test	Mother	07/08/2020, 14:11:23	Linear	0.00 GB		

Figure 56: Impersonate user and navigate on his dashboard temporarily



MAON Documentation Dashboard Nicolai Schmid

Planned maintenance on 2025/09/08 from 12am to 3pm

Quota 1 nms Slots 0/2 nms Storage 5 GB/250 GB

Projects Create Column visibility Show filters Reset view

Project name	Scenario count	Finished simulations count	Running simulations count	Pending simulations count	Storage usage
Upload Test	3	0	0	0	0.00 GB
Test	2	2	0	1	0.00 GB
Backtesting	4	0	0	0	0.00 GB

3 3 Rows 1 of 1

Simulations Column visibility Show filters Reset view

Type	Status	Project name	Scenario name	Submit time	Mode	Storage usage	Output	Delete
Single	Running	Test	Mother	07/08/2020, 14:11:24	Linear	0.00 GB		
Single	Failed	Test	Mother	07/08/2020, 14:11:23	Linear	0.00 GB		

Tasks Liable simulations Platform settings Changelog Profile Stop Impersonation

Figure 57: Stop impersonation and return to admin dashboard

MAON

Documentation Dashboard Admin Fabian Breitkreutz

Create

Column visibility Show filters Reset view

Email	First Name	Last Name	Status	Storage Usage (GB)	Signed up On	Running Simulations	Status Action	Edit	Impersonate	Export
mk@maon.eu	Mihail	Kulov	active	0	16/11/2019, 11:27:06	0	Deactivate		Impersonate	Export table simulations
so@maon.eu	Nicolai	Schmid	active	0	15/01/2020, 10:11:17	0	Deactivate		Impersonate	Export table simulations
info@maon.eu	Mihail	Kulov	active	0	31/10/2019, 11:29:42	0	Deactivate		Impersonate	Export table simulations
kim.taylor@p3-group...	Kim	Taylor	active	0	17/11/2019, 11:47:28	0	Deactivate		Impersonate	Export table simulations
inwoker@mail.de	Chensen	Wang	active	0	16/01/2020, 10:27:05	0	Activate		Impersonate	Export table simulations
s7sara@gmail.com	Saravanan	Shanmugam	active	0	17/01/2020, 09:14:01	0	Deactivate		Impersonate	Export table simulations
fb@maon.eu	Fabian	Breitkreutz	active	0.01	12/01/2020, 12:17:21	0	Deactivate		Impersonate	Export table simulations
thomas.anderski@a...	Thomas	Anderski	active	0	24/01/2020, 14:09:09	0	Activate		Impersonate	Export table simulations
hz@maon.eu	HuangJolun	Zhou	active	0	20/08/2019, 13:15:13	0	Activate		Impersonate	Export table simulations
henrik.sommer@tu-ill...	Henrik	Sommer	active	0	23/08/2019, 14:14:25	0	Activate		Impersonate	Export table simulations

10 159 Rows

1 of 16

Figure 58: Return to the admin panel after stopping impersonation

Module 4

• Maon's Core applications Testing

This module is a bit separate from Maon's web application. It uses python and shell to test and log Maon's Core application. This application consumes inputs from a file and generates outputs that will be used in the web application simulations.

- Inputs: The input data can be changed in multiple interfaces such as demands and batteries datasets. The configuration of the technical setup of the simulation procedure can be changed in the project configuration.
- Outputs: the output data determines the results of the processing of the simulations created. They can be devised into categories as they maintain certain files that are crucial for the results we're deriving from the platform.
- Core application: this is the main application that simulates energies and is considered the main product of Maon services. The platform helps create it and it should be tested for the optimization and durability of the product.

```

$ python -m venv venv
$ source venv/bin/activate
$ pip install -r requirements.txt
Requirement already satisfied: polars in ./venv/lib/python3.13/site-packages (from -r requirements.txt (line 1)) (1.29.0)
Requirement already satisfied: python-dotenv in ./venv/lib/python3.13/site-packages (from -r requirements.txt (line 2)) (1.1.0)

[notice] A new release of pip is available: 25.1.1 -> 25.2
[notice] To update, run: pip install --upgrade pip
(venv) test $ cd helpers
(venv) helpers $ ls
conf:cpp_snlppet* converter.py
(venv) helpers $ python convert.py
python: can't open file '/home/bacen/WORKSPACE/test/helpers/convert.py': [Errno 2] No such file or directory
(venv) helpers $ python converter.py
[WARN] Line 399 did not match: CConfVar<double> CConf::dThPwrCostFuelPerGjMax = CConfVar<double>("thermal_power_cost_fuel_per_gj_maximum", 1000, 0, dNix - 2);
[WARN] Line 403 did not match: CConfVar<double> CConf::dThPwrCostTransPerGjMax = CConfVar<double>("thermal_power_cost_transport_per_gj_maximum", 1000, 0, dNix - 2);
[WARN] Line 416 did not match: CConfVar<double> CConf::dThPwrRestrictEmissSlackCostMaxOverstep = CConfVar<double>("thermal_power_restriction_emission_slack_cost_maximum_overstep", 1000, 0, dNix - 2);
[WARN] Line 417 did not match: CConfVar<double> CConf::dThPwrRestrictEmissSlackCostMinUnderstep = CConfVar<double>("thermal_power_restriction_emission_slack_cost_minimum_understep", 1000, 0, dNix - 2);
[WARN] Line 428 did not match: CConfVar<double> CConf::dThPwrSpecificTco2PerGjMax = CConfVar<double>("thermal_power_specific_tco2_per_gj_maximum", 10000, 0, dNix - 2);
[WARN] Line 755 did not match: CConfVar<unsigned> CConf::uDrShiftMax = CConfVar<unsigned>("demand_side_response_shift_maximum", 24, 0, dNix - 2);
[WARN] Line 774 did not match: CConfVar<unsigned> CConf::uProTotParallelThreads = CConfVar<unsigned>("procedure_total_parallel_threads", max(1, std::thread::hardware_concurrency()), 1, 1024);
[WARN] Line 775 did not match: CConfVar<unsigned> CConf::uProTotWorkingMemoryMb = CConfVar<unsigned>("procedure_total_working_memory_mb", getRAMSizeMB() + getSWAPSizeMB(), 1, 4194304);
[WARN] Line 777 did not match: CConfVar<unsigned> CConf::uSolverMIPcpuTimeLimitStartFirst = CConfVar<unsigned>("solver_mixed_integer_problem_cpu_time_limit_start_first", (120 * 40), 1, 43200);
[WARN] Line 778 did not match: CConfVar<unsigned> CConf::uSolverMIPcpuTimeLimitStartSecond = CConfVar<unsigned>("solver_mixed_integer_problem_cpu_time_limit_start_second", (240 * 40), 1, 43200);
done! Check 'configurations.csv'

```

Figure 59: Converting cpp files to csv files

```
(venv) test $ python main.py configuration_tests
Tested core app: /home/bacem/Documents/maon
Running configuration_tests. Please wait ...
Tests complete! ✓
```

subtype/type	success	fail
---	---	---
str	str	str
conf overstep	505	0
conf understep	505	0
conf max	505	0
conf min	505	0
-----	-----	-----
unit	0	0
-----	-----	-----
integration	0	0
-----	-----	-----
Exception	310	0
-----	-----	-----
sum	2330	0

Figure 60: Consume csv files and run configuration tests

The multiple results shown are by testing the inputs, we could define them as follows:

- Configuration overstep: determines when the value is expected to exceed the maximum value and it does.
- Configuration understep: determines when the value is expected to be lower than the minimum value and it does.
- Configuration max: determines when the value is expected to equal the maximum value and it does.
- Configuration min: determines when the value is expected to equal the minimum value and it does.
- Unit tests: Unit tests run for the simulation application that helps test and predict the functionality of the core application.
- Integration tests: determine the combination of different integrations for the core application and its usability for further needs.
- Exception: inputs that don't align with the criteria such as type are stored in a csv file and counted through the tests.

Errors sometimes can be displayed if there's a malfunction which makes it easier for the tester to know which inputs are corrupted and needs fixes. We can display the errors shown in the next figure as follows:

- Error in hydro_slack_cost_filling_end_deficit: Set value 99999999.00000001 above the maximum valid value 99999999.0 and thus expected to fail, but app runs through. (Double type)

- Error in `hydro_slack_cost_filling_end_surplus`: Set value `99999999.000000001` above the maximum valid value `99999999.0` and thus expected to fail, but app runs through. (Double type)

```
$ python main.py configuration_tests
Tested core app: /home/bacem/Documents/maon
Running config tests. Please wait ...

Error in hydro_slack_cost_filling_end_deficit: Set value 99999999.000000001 above the maximum
Error in hydro_slack_cost_filling_end_surplus: Set value 99999999.000000001 above the maximum

Tests complete!
```

subtype/type	success	fail
conf overstep	493	2
conf understep	495	0
conf max	495	0
conf min	495	0
unit	0	0
integration	0	0
Exception	331	0
sum	2309	2

Figure 61: Run configuration tests while there are errors

```
(venv) test $ python main.py unit_tests
Tested core app: /home/bacem/Documents/maon
Running unit_tests. Please wait ...
Tests complete! ✓
```

subtype/type	success	fail
conf overstep	0	0
conf understep	0	0
conf max	0	0
conf min	0	0
unit	45	0
integration	0	0
Exception	0	0
sum	45	0

```
(venv) test $
```

Figure 62: Run unit tests

V. Conclusion

This chapter explores the various contributions made to the Maon web application, highlighting its evolution from a buggy and partially incompatible platform to a more stable, efficient, and user-friendly system following the internship. The improvements significantly enhanced the user experience and strengthened the platform's ability to fulfill its mission and attract new users. Several key contributions were made, including bulk data handling, leap year calculations, data processing, and energy simulation testing. Additional enhancements included the integration of the DSR-to-map feature, persistent filters during navigation, web scraping functionalities, corrections to inherited child display issues, hydro color adjustments, and other optimizations.

General conclusion

During this six-month internship at MAON GmbH, I had the opportunity to contribute to the maintenance and continuous improvement of the company's web application dedicated to energy simulation and analysis. Working within a professional environment allowed me to deepen my understanding of full-stack development using the MERN stack (MongoDB, Express.js, React, and Node.js) while gaining valuable experience in testing and validating energy simulation processes with Python.

Throughout the project, I focused on multiple categories that were mentioned in table 1, we can mention the following:

- **Documentation & Logging.**
- **UI/UX Stability.**
- **Frontend–Backend Integration.**
- **Alert System.**
- **Energy Monitoring.**

My contribution came to optimizing the application's performance, fixing bugs, implementing new features, and ensuring smooth integration between the front-end and back-end systems. I also participated in testing simulation modules, which enhanced my analytical and problem-solving skills, as I learned to identify and address potential issues affecting the accuracy and efficiency of simulations. This experience strengthened my technical competencies, especially in web application maintenance, data handling, and software testing, while also improving my ability to collaborate within a multidisciplinary and international team.

Overall, this project was an enriching and formative experience that deepened my expertise in modern web technologies and their applications in the energy simulation sector. It also confirmed my motivation to continue developing solutions that contribute to the advancement of sustainable and intelligent energy systems.

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